

A

MAJOR PROJECT REPORT ON

IoT ENABLED SAFETY HELMET FOR COAL MINES

Submitted in partial fulfilment of the requirements for the award of the degree

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

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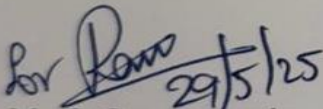
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CERTIFICATE

This is to certify that the major project entitled "**IoT ENABLED SAFETY HELMET FOR COAL MINES**" being submitted by (S. Srinivas 211FA05017), (N. Divyasri 211FA05018), (V. Pavani 211FA05054), (Y. Naga Sai 211FA05260) respectively, in partial fulfilment of the requirements for the award of the B. Tech degree in Electronics and Communication Engineering at Vignan's Foundation for Science, Technology & Research, is a record of original work carried out by us under the guidance of **Mr.G.Ganga prasad** (Assistant professor).


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DECLARATION

We hereby declare that our project work described in the major project titled "**IoT ENABLED SAFETY HELMET FOR COAL MINES**" which is being submitted by us for the partial fulfilment in the department of ECE, Vignan's Foundation for Science, Technology and Research (Deemed to be University), Vadlamudi, Guntur, Andhra Pradesh, and the result of investigations are carried out by us under the guidance of **Mr.G.Ganga prasad**

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Finally, we would like to thank our parents and friends for the moral support throughout the project work.

With Sincere Regards,

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Project Title	IoT ENABLED SAFETY HELMET FOR COAL MINE
Programme Concentration Area	IoT(Internet of Things)

Constraints: Examples	
Economic	Cost-effective components to ensure affordability for widespread implementation
Environmental	Materials used should be recyclable and comply with e-waste regulations.
Sustainability	It should use low power or renewable energy sources like solar charging.
Manufacturability	The design must support easy and cost-effective mass production
Ethical	collected data must respect worker privacy and not be misused
Health and Safety	Included gas and temperature sensors
Social	Promote safety
Political	Supports government mandates on labor safety in hazardous environments
Others	Includes features like SOS alert, fall detection, and wireless communication

Standard/Tool	Usage
IEEE 1451	Enables integration and communication of smart sensors in the helmet.
IEEE 802.15.4	Provides low-power wireless communication suitable for underground environments.
IEEE 11073	Enables transmission of health monitoring data from wearable sensors.

Previous Course Required for Major Project	Internet of Things (IoT), Microcontrollers and Applications, Sensors and Transducers, Wireless Communication, Industrial Safety Engineering, C/Embedded C Programming
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Major Project SDG Mapping

SDG Number	SDG Title	Project Relevance
1	No Poverty	NA
2	Zero Hunger	NA
3	Good Health and Well-being	Enhances worker safety and health monitoring in hazardous mining environments
4	Quality Education	NA
5	Gender Equality	NA
6	Clean Water and Sanitation	NA
7	Affordable and Clean Energy	NA
8	Decent Work and Economic Growth	Promotes safer working conditions, contributing to improved labor standards
9	Industry, Innovation and Infrastructure	Applies IoT and embedded systems to improve industrial safety practices
10	Reduced Inequality	NA
11	Sustainable Cities and Communities	Helps build safer, resilient mining infrastructure supporting sustainable communities
12	Responsible Consumption and Production	NA
13	Climate Action	Supports environmental monitoring in mines to reduce harmful emissions and hazards
14	Life Below Water	NA
15	Life on Land	NA
16	Peace, Justice and Strong Institutions	NA
17	Partnerships for the Goals	NA

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ABSTRACT

Coal mining remains one of the most hazardous occupations due to the presence of toxic gases, high temperatures, poor visibility, and the constant threat of structural collapse. Ensuring the safety of miners in such environments is of paramount importance. This project presents the design and development of an IoT-enabled smart helmet aimed at enhancing safety and communication in underground coal mines. The proposed helmet is equipped with multiple sensors to continuously monitor environmental parameters such as temperature, humidity, methane gas concentration, and carbon monoxide levels. The helmet also features an accelerometer to detect falls or sudden impacts that may indicate accidents. The collected data is transmitted in real-time to a central monitoring system using wireless communication technologies, enabling prompt alerts and rapid response to hazardous conditions. The integration of Internet of Things (IoT) technology facilitates remote monitoring and data logging, thereby improving decision-making and preventive measures. This smart helmet not only enhances the safety of miners but also contributes to the advancement of intelligent mining infrastructure.

CHAPTER 1

INTRODUCTION

1.1 Background

Coal mining is a cornerstone of the global energy supply chain, particularly in developing countries where it supports electricity generation and industrial operations. However, the mining industry is also associated with some of the highest occupational hazards. Workers often operate in deep underground tunnels, where environmental conditions can change rapidly and without warning. Hazards such as methane explosions, carbon monoxide poisoning, landslides, equipment-related injuries, and prolonged exposure to heat and dust are constant threats to miners' lives. Traditional safety approaches—such as protective clothing, fixed-location gas sensors, and manual inspection—offer limited protection and often fail to provide real-time, individualized hazard detection.

1.2. Need for Smart Safety Solutions

The increasing demand for safer mining operations has prompted researchers and engineers to look beyond conventional methods. Current helmet designs are passive, offering physical protection against impact but lacking the ability to detect environmental or physiological threats. Furthermore, most monitoring equipment used in mines is either bulky, localized, or disconnected from the workforce. These limitations result in delayed response times during emergencies. There is a clear need for a smarter, integrated solution that provides miners with personal, mobile, and real-time safety monitoring—one that can detect issues before they become life-threatening.

1.3. Role of IoT in Mining Safety

The Internet of Things (IoT) introduces a paradigm shift in industrial safety systems by enabling real-time, wireless communication between physical devices and centralized monitoring platforms. In the context of mining, IoT facilitates the development of wearable devices—like smart safety helmets—that can collect and transmit data related to environmental conditions, miner health, and physical movements. These helmets typically include:

- **Gas sensors** (e.g., MQ-2, MQ-135) for detecting hazardous gases such as CH₄ (methane), CO (carbon monoxide), and CO₂.
- **Temperature and humidity sensors** (like DHT11 or DHT22) to monitor climate conditions that could lead to heat stress or fire risks.
- **Accelerometers** (like ADXL345) for fall or impact detection.
- **Pulse or heart rate sensors** to monitor the physical state of the worker.
- **Communication modules** such as GSM, Wi-Fi, Zigbee, or LoRa to send real-time alerts and data to the surface-level monitoring system.

This combination of sensors and wireless communication allows for timely detection of anomalies, enabling faster response and improved safety outcomes.

1.3.1 Importance of smart helmet in coal mines

The importance of smart helmets in coal mines lies in their ability to enhance safety, efficiency, and monitoring capabilities in one of the most hazardous working environments. Traditional mining helmets provide only physical protection, while smart helmets integrate sensors, communication modules, and IoT technology to offer real-time situational awareness and emergency response capabilities. Below are key points highlighting their importance:

1. Real-Time Monitoring of Hazardous Gases

- Smart helmets are equipped with sensors to detect the presence of toxic gases like methane (CH₄), carbon monoxide (CO), and hydrogen sulfide (H₂S).
- This real-time detection helps in early warning and evacuation, thus reducing the chances of gas explosions and suffocation incidents.

2. Worker Tracking and Location Awareness

- GPS or RFID modules allow real-time tracking of workers underground.
- In case of accidents such as collapses or gas leaks, rescue operations can be more targeted and efficient.

3. Fall and Collision Detection

- Accelerometers and gyroscopes in smart helmets can detect falls, impacts, or abnormal motion.
- The system can trigger immediate alerts to the control room or supervisors for quick response.

4. Health Monitoring

- Some smart helmets include sensors to monitor heart rate, body temperature, and fatigue levels, helping to prevent incidents due to worker exhaustion or health issues.

5. Two-Way Communication

- Smart helmets often support wireless communication (Bluetooth, Zigbee, LoRa, etc.), enabling workers to report issues and receive instructions without delay.

6. Data Logging and Analytics

- Continuous data collection helps in risk assessment, regulatory compliance, and predictive maintenance of mine infrastructure.

7. Improved Safety Compliance

- Automated alerts and logs help ensure workers follow safety protocols, reducing human error and non-compliance.

1.4. Objective of the System

The primary objective of the IoT-enabled safety helmet is to enhance miner safety and reduce fatalities by providing a **comprehensive, mobile, and intelligent safety monitoring system**. Key goals include:

- Real-time detection of hazardous gas concentrations.
- Monitoring temperature and humidity for early fire or heat warning.
- Fall and motion detection to identify accidents or health emergencies.
- Wireless communication of alerts and data to a centralized dashboard or mobile app.
- Location tracking to support rescue operations during emergencies.

The system also aims to reduce the burden on manual inspection processes and improve operational efficiency by offering data-driven insights into mine conditions and worker safety trends.

1.5. Significance and Scope

The integration of IoT in personal protective equipment, especially in mining, marks a significant innovation in industrial safety. The IoT-enabled helmet transforms a simple protective tool into a **smart wearable safety system**. The significance of this solution includes:

- **Personalized safety:** Each miner can be monitored individually, improving responsiveness in emergencies.
- **Data centralization:** Supervisors can track multiple helmets/workers from a single dashboard.
- **Predictive maintenance:** Historical data can be used to identify patterns and prevent future incidents.
- **Scalability:** The helmet system can be scaled and customized for different mine environments or industries.

Moreover, such systems can be integrated into a larger **Industrial IoT (IIoT)** network, enabling coordinated safety and productivity monitoring across an entire mining operation. This system also supports regulatory compliance, insurance verification, and workforce training based on real-time data and incident logs.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to Smart Safety helmet in Mining

The mining industry has long grappled with occupational hazards such as gas leaks, rock falls, and extreme environmental conditions. Traditional safety systems, although valuable, are often static, reactive, and offer limited protection in rapidly evolving scenarios. This has led researchers and engineers to explore advanced technologies like the Internet of Things (IoT) to enhance worker safety in underground mines.

2.2 Role of IoT in Personal Protective Equipment (PPE)

IoT-enabled safety devices transform standard PPE—like helmets and jackets—into intelligent monitoring tools. These devices incorporate embedded sensors, wireless communication modules, and microcontrollers to track environmental and physiological parameters in real-time. Smart helmets, in particular, have emerged as a promising wearable solution for high-risk industries such as mining.

2.3 Key Parameters Monitored in Smart Helmets

Most of the existing literature focuses on the integration of sensors for:

- **Gas detection** (e.g., methane, carbon monoxide, and ammonia)
- **Temperature and humidity monitoring**
- **Fall and motion detection**
- **Heart rate or pulse monitoring**
- **Location tracking via GPS**

The collected data is transmitted through wireless technologies such as GSM, Zigbee, LoRa, and Wi-Fi to a centralized control system or cloud platform for real-time monitoring and emergency response.

2.4 Literature survey

- **“Energy Harvesting Methods for Wearable Mining Safety Devices”**

Authors: Rodriguez, Kim(2023)

Main Aim: To explore and evaluate various energy harvesting methods—kinetic, thermal, and photovoltaic—suitable for powering helmet electronics in underground mining environments, thereby extending their operational time without battery replacement.

Limitations:

- * Photovoltaic harvesting is limited due to low light levels underground.
- * Kinetic and thermal methods may not generate sufficient energy for high-power components.
- * Integration complexity and cost could hinder practical deployment.

- **“Machine Learning Applications in Mining Safety Systems”**

Authors: Chen, et al.(2021)

Main Aim: To develop machine learning algorithms capable of predicting hazardous mining conditions before they become critical, enabling integration into helmet-based early warning systems.

Limitations:

- * Requires high-quality, continuous sensor data for accurate prediction.
- * Potential false positives or missed alerts in unpredictable scenarios.
- * Deployment complexity in real-time underground environments.

- **“Thermal Management in Underground Mining Helmets Passive and Active Cooling Systems Evaluation”**

Authors: Ramakrishnan, et al.(2021)

Main Aim: To compare passive and active cooling strategies in mining helmets and determine the most effective method for reducing heat stress without compromising helmet integrity.

Limitations:

- * Phase-change materials add weight and cost.
- * Passive systems may not suffice in extremely high temperatures.
- * Long-term durability and maintenance of cooling systems not fully tested.

- **“Finite Element Analysis of Impact Energy Absorption in Mining Safety Helmets”**

Authors: Liu, et al.(2020)

Main Aim: To use finite element analysis (FEA) to study how different helmet designs absorb impact energy, identifying optimal shell thickness and suspension systems to improve protection against falling objects.

Limitations:

- * Simulated models may not fully capture real-world impact variability.
- * Material behavior under dynamic conditions needs experimental validation.
- * FEA findings require practical testing before implementation.

- **“Testing Methodologies for Next-Generation Mining PPE Certification”**

Authors: Chang, DuPont (2020)

Main Aim: To propose updated certification and testing standards for smart mining helmets, focusing on sensor accuracy and electronic system durability not covered by traditional protocols.

Limitations:

- * Proposed methods require new regulatory adoption and validation.
- * Implementation may involve high testing costs and extended timelines.
- * May not account for all extreme mining conditions.

- **“Self-Healing Polymer Applications in Industrial Safety Equipment”**

Authors: Nakamura, Singh (2020)

Main Aim: To investigate self-healing polymer materials that automatically repair minor structural damage in mining helmets, improving longevity and safety.

Limitations:

- * Limited to minor impact recovery; may not handle severe damage.
- * Long-term material behavior under mining stress needs further study.
- * High production cost of self-healing polymers.

- **“Integrated Gas Detection Systems in Smart Mining Helmets: Field Testing Results”**

Authors: Wang, Chen (2019)

Main Aim:

To evaluate the accuracy and responsiveness of embedded gas sensors (methane and CO) in smart helmets during real-world underground mining conditions.

Limitations:

- * Sensor calibration can be affected by humidity and dust.
- * Limited lifespan of miniaturized sensors.
- * Data transmission reliability underground not fully addressed.

- **“Comparative Analysis of International Mining Head Protection Standards”**

Authors: Harrison, et al. (2019)

Main Aim: To compare mining helmet safety standards globally and identify inconsistencies, recommending unified certification practices for easier international compliance.

Limitations:

- * Global harmonization is difficult due to regional policy differences.
- * Some standards may resist change due to legacy systems.
- * Recommendations need extensive stakeholder consensus.

CHAPTER 3

EXISTING SYSTEM

3.1 Traditional helmet and importance of Smart Helmet

Traditional Helmet in Coal Mines

Traditional helmets have long been a basic safety requirement in coal mines. Their primary purpose is to protect workers from physical injuries due to falling rocks, low overhead surfaces, or equipment-related accidents. These helmets are typically made from hard materials like high-density polyethylene (HDPE) and offer a durable, lightweight solution for head protection. However, they provide only passive safety—offering no capabilities for environmental sensing, health monitoring, or communication. As such, while they are essential, they fall short in addressing the complex and dynamic risks present in underground mining environments.

Importance of Smart Helmet

Smart helmets are an advanced evolution of traditional helmets, incorporating Internet of Things (IoT) technologies and sensor-based systems. These helmets are equipped with gas sensors that continuously monitor the air for harmful gases, alerting workers and control centers in real time. They also feature GPS or RFID tracking systems that help locate miners quickly in the event of accidents such as cave-ins or gas leaks.

3.2 The following are the Existing Smart Helmets

3.2.1 Smart Helmet Using Arduino and Gas Sensors

- Description: A basic IoT-enabled helmet prototype using an Arduino microcontroller, gas sensors (MQ-series), and a temperature sensor to monitor the environment.
- Features:
 - Detects harmful gases (like CO, CH₄).
 - Monitors temperature.
 - Sends alerts via Wi-Fi or GSM modules.
- Limitations:
 - Limited range and no real-time location tracking.
 - Not industrial-grade or ruggedized for mining environments.

3.2.2 Helmet with ZigBee-Based Wireless Communication

- Description: Utilizes ZigBee modules for low-power, short-range communication. Sensors monitor hazardous gas levels and ambient temperature.

- Features:
 - Data is transmitted to a control room wirelessly.
 - Basic alert mechanisms.
- Limitations:
 - ZigBee has limited range and is not suitable for large-scale mines.
 - Lack of GPS tracking or impact detection.

3.2.3 GSM/GPRS-Based Safety Helmet for Remote Monitoring

- Description: Incorporates GSM/GPRS for long-distance communication between the miner's helmet and the base station.
- Features:
 - Real-time transmission of environmental data.
 - SMS alerts to supervisors.
- Limitations:
 - GSM signals may be unreliable in underground mines.
 - Limited sensor integration (usually just 2–3 sensors).

3.2.4 IoT-Based Helmet with GPS and Fall Detection

- Description: Advanced systems that combine environmental sensors with accelerometers and GPS modules.
- Features:
 - Tracks miner's location.
 - Detects falls or collisions.
 - Sends automatic alerts to rescue teams.
- Limitations:
 - GPS is ineffective underground unless used with hybrid positioning systems.
 - Costlier and power-intensive.

3.2.5 LoRa-Based Smart Helmet System

- Description: Uses LoRa (Long Range) communication protocol ideal for underground or remote environments.
- Features:
 - Long-distance, low-power data transmission.

- Environmental monitoring (gas, humidity, temperature).
- LoRa gateways collect and forward data to the cloud.
- Limitations:
 - Lower data rate than Wi-Fi.
 - Requires proper gateway placement for optimal performance.

Overall limitations of traditional helmets

3.3 Overall Limitations of Traditional Helmets in Coal Mines

Traditional helmets, while essential for basic head protection, have several limitations when it comes to ensuring complete safety in hazardous environments like coal mines. These limitations make them insufficient for addressing the complex and dynamic risks faced by miners today.

1. basic head protection
2. Lack environmental sensing
3. Health monitoring, communication
4. tracking and data logging capabilities essential for modern coal mine safety.

CHAPTER 4

PROPOSED SYSTEM

The proposed system, illustrated the smart helmet which consists of transmitter and receiver section. The system consists of integrating multiple sensors and communication modules into a single unit powered by an Arduino microcontroller. The system includes a vibration sensor, CO (carbon monoxide) sensor, DHT11 (for temperature and humidity), and a MEMS-based fall detection sensor. These sensors continuously monitor the environment and the user's physical condition. Data from these sensors is processed by the Arduino, which then transmits critical alerts through a Zigbee transmitter for remote monitoring. Simultaneously, real-time data is displayed on an LCD screen, and an audible alert is triggered via a buzzer in case of anomalies. A regulated power supply ensures the stable operation of all components. This integrated and automated setup significantly enhances safety monitoring, real-time alerting, and communication efficiency, making it ideal for applications such as elderly care, worker safety, or environmental monitoring

4.1 Zigbee Transmitter (Tx)



fig.4.1.1 Transmitter section

4.1.1.1 Description of Zigbee Transmitter (Tx) –

In the Arduino-Based Transmitter Unit

Function:

- The Zigbee Transmitter (Tx) module is connected to the Arduino microcontroller.
- It is responsible for wirelessly sending sensor data (from vibration, CO, temperature, humidity, and fall detection sensors) to a remote device.
- Operates on IEEE 802.15.4 standard, designed for low-power and low-data-rate communication, ideal for sensor networks.

Role in the System:

- Collects processed data from the Arduino and transmits it to the Zigbee Receiver connected to the ESP32 in the remote monitoring unit.

- Ensures real-time and wireless data transfer, eliminating the need for wired communication between distant units.

4.2 Zigbee Receiver (Rx)

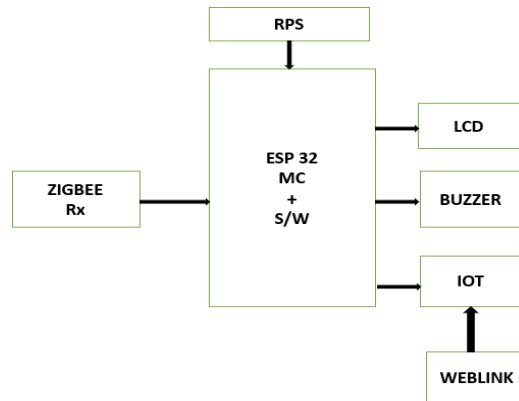


fig.4.2.1 Transmitter section

4.2.1.1 Description Zigbee Receiver (Rx) –

In the ESP32-Based Receiver Unit

Function:

- The Zigbee Receiver (Rx) module is connected to the ESP32 microcontroller.
- It receives the sensor data wirelessly from the Zigbee Transmitter.
- Acts as the entry point for all data coming from the field sensor unit (Arduino system).

Role in the System:

- Passes the received data to the ESP32 for further processing.
- The ESP32 then:
 - Displays the data on the LCD,
 - Activates alerts through the buzzer if needed,
 - Sends the data to an IoT platform for cloud-based monitoring.

This Zigbee link enables **wireless, low-cost, and reliable data transfer** between the sensor node and the remote monitoring unit, making it ideal for applications like industrial safety, healthcare, or home automation.

4.3 Circuit diagrams of receiver section

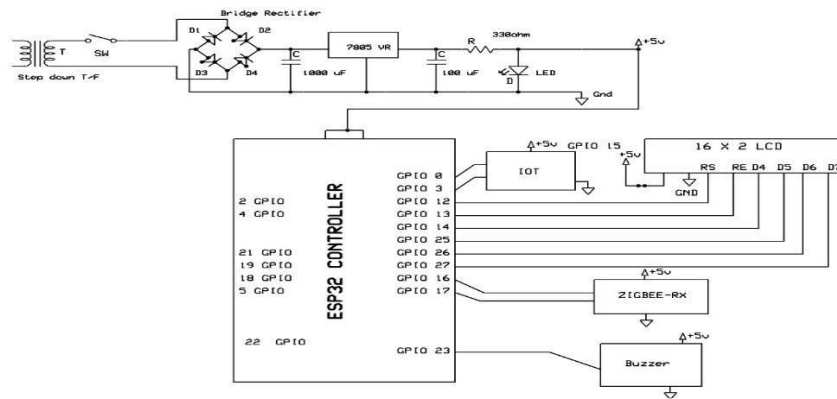


fig.4.3.1 esp32 controller

Here's a detailed explanation of the **receiver circuit diagram** using an **ESP32 controller** for the coal mine safety helmet system:

4.3.1.1 DECSRIPTION OF RECEIVER SECTION

Power Supply Section

- **Step-down Transformer:** Reduces AC mains voltage (typically 230V) to a lower AC voltage.
- **Bridge Rectifier (D1–D4):** Converts AC to pulsating DC.
- **Filter Capacitors (1000µF & 100µF):** Smoothens the DC signal by reducing ripple.
- **Voltage Regulator (7805):** Regulates output to a constant +5V DC.
- **Indicator LED (with series resistor):** Confirms power is available.

This 5V output powers the ESP32 and all connected modules (LCD, Zigbee, buzzer, etc.).

ESP32 Microcontroller Section

- Acts as the **main controller** in the receiver unit.
- Interfaces with:
 - Zigbee Receiver Module
 - IoT Module
 - 16x2 LCD Display
 - Buzzer

ESP32 GPIO pins are used to receive signals and control output devices.

Zigbee-RX Module

- Connected to **GPIO 23** of ESP32.
- Receives data wirelessly from the transmitter (helmet unit with sensors).

- Sends this data to the ESP32 for further processing and display.

IoT Module

- Connected to **GPIO 15**.
- Used to send received sensor data to a **cloud platform** or remote server for real-time monitoring.
- Enables remote access to data, improving miner safety management.

16x2 LCD Display

- Connected to GPIOs (likely GPIOs 12 to 14, 25–27 based on typical LCD pin connections).
- Displays sensor values and alerts in real-time..

Buzzer Module

- Connected to **GPIO 17**.
- Triggers an **audible alert** in case of any hazardous condition detected (like gas leak or fall).

Power Connections

- All modules (Zigbee, LCD, IoT, Buzzer) are powered by **regulated +5V** from the power supply section.
- Common ground is connected across all components to complete the circuit.

4.4 Circuit diagrams of Transmitter section

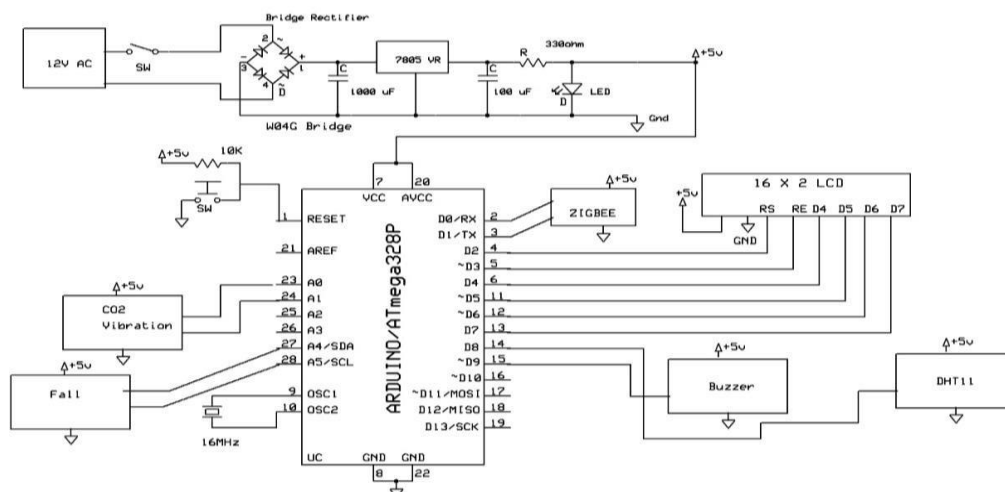


fig.4.4.1 ARDUINO/Atmega328P

4.4.1.1 Description of receiver section

Power Supply Section:

1. 230V AC Input (top-left corner):

- This is the main AC power supply source.
 - Enters the circuit to be converted into DC voltage.
2. **Step-Down Transformer (12V AC):**
- Converts 230V AC to 12V AC.
 - Suitable for low-voltage electronics.
3. **Bridge Rectifier (Using 1N4007 Diodes):**
- Converts AC to pulsating DC.
 - Made of four diodes in a bridge configuration.
4. **Filter Capacitor (1000 μ F/25V):**
- Smooths out the DC from the rectifier.
 - Removes ripple.
5. **Voltage Regulator (7805 IC):**
- Regulates the voltage down to +5V DC.
 - Powers 5V logic circuits.
6. **LED with 330 Ω Resistor:**
- Indicator to show power is ON.

Microcontroller Section:

7. **Arduino ATmega328P:**
- Main microcontroller.
 - Controls the logic and communication between components.
 - Pins:
 - Analog inputs (A0–A5): Used for sensors like CO2 and Vibration.
 - Digital I/O (D0–D13): Used for modules like LCD, Zigbee, buzzer, etc.
8. **16 MHz Crystal Oscillator + Capacitors:**
- Provides the necessary clock for ATmega328P operation.
 - Helps in accurate timing.

Sensor Modules:

9. **CO2 Sensor:**
- Measures the concentration of CO2 in the air.
10. **Vibration Sensor:**

- Detects vibration or movement.
- Helps detect machinery operation or structural instability.

11. Fall Detection Sensor:

- Detects if a person wearing the helmet has fallen.
- Could be an accelerometer or tilt switch.

12. DHT11 Sensor:

- Measures temperature and humidity.
- Connected to a digital pin.

Output Devices:

13. 16x2 LCD Display:

- Displays environmental data like CO₂, temperature, humidity, fall detection.
- Connected in 4-bit mode using pins RS, E, D4–D7.

14. Buzzer:

- Audible alert for hazardous conditions.
- Triggered by the microcontroller.

Communication Module:

15. Zigbee Module:

- Enables wireless communication with a central monitoring station.
- Uses serial communication (TX/RX pins).

CHAPTER 5

HARDWARE DESCRIPTION

The following are the hardware sections of this project

5.1 Introduction to Microcontrollers:



Fig.5.1.1 microcontroller

Circumstances that we find ourselves in today in the field of microcontrollers had their beginnings in the development of technology of integrated circuits. This development has made it possible to store hundreds of thousands of transistors into one chip. That was a prerequisite for production of microprocessors, and the first computers were made by adding external peripherals such as memory, input-output lines, timers and other. Further increasing of the volume of the package resulted in creation of integrated circuits. These integrated circuits contained both processor and peripherals. That is how the first chip containing a microcomputer, or what would later be known as a microcontroller came about.

Microprocessors and microcontrollers are widely used in embedded systems products. Microcontroller is a programmable device. A microcontroller has a CPU in addition to a fixed amount of RAM, ROM, I/O ports and a timer embedded all on a single chip. The fixed amount of on-chip ROM, RAM and number of I/O ports in microcontrollers makes them ideal for many applications in which cost and space are critical. The AVR is a modified Harvard architecture 8-bit RISC single chip microcontroller which was developed by Atmel in 1996. The AVR was one of the first microcontroller families to use on-chip flash memory for program storage, as opposed to One-Time Programmable ROM, EPROM, or EEPROM used by other microcontrollers at the time.

5.2 Arduino uno

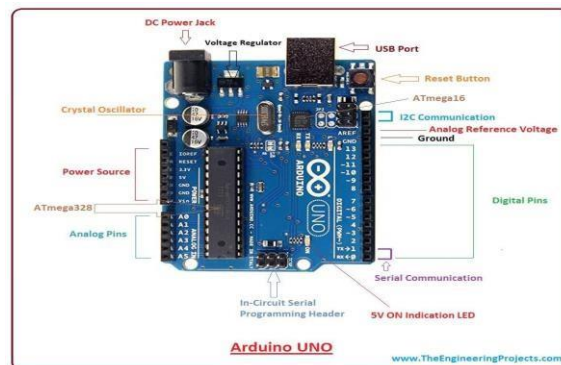


fig.5.2.1 pin diagram of atmega328

5.2.2 Explanation of pin diagram

- DC Power Jack: Connects external 7–12V DC power supply to power the board.
- Voltage Regulator: Converts input voltage to a stable 5V output for the board.
- USB Port: Used to program the board and provide 5V USB power.
- Reset Button: Resets the microcontroller, restarting the program.
- ATmega328: Main microcontroller that runs the user's code.
- ATmega16 (USB-to-Serial): Converts USB data to serial for ATmega328 communication.
- Crystal Oscillator: Provides a 16 MHz clock signal for precise timing.
- Power Source Pins (3.3V, 5V, GND, Vin): Provide power outputs and accept external voltage input.
- Analog Pins (A0–A5): Read analog sensor values (0–1023 range).
- Digital Pins (0–13): Used for digital input/output (HIGH or LOW).
- I2C Communication (A4 SDA, A5 SCL): Supports I2C protocol for communication with other I2C devices.
- Serial Communication (TX, RX - Pins 0 and 1): Enables communication with computers or other serial devices.
- Analog Reference Voltage (AREF): Used to set a reference voltage for analog input.
- Ground (GND): Common reference point for all electrical components.
- In-Circuit Serial Programming Header (ICSP): Used for burning bootloader or firmware onto the ATmega chip.
- 5V ON Indication LED: Lights up when board is powered on.

5.3 ESP32 CONTROLLER

ESP32 is the SoC (System on Chip) microcontroller which has gained massive popularity recently. Whether the popularity of ESP32 grew because of the growth of IoT or whether IoT grew because of the introduction of ESP32 is debatable. If you know 10 people who have been part of the firmware development for any IoT device, chances are that 7–8 of them would have worked on ESP32 at some point.

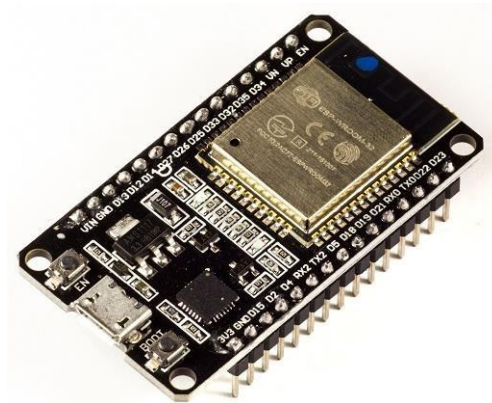


fig.5.3.1 esp32 controller

The ESP32 is a powerful, low-cost microcontroller developed by Espressif Systems, widely used in IoT, embedded systems, and wireless communication projects. It is the successor to the popular ESP8266 and offers significant improvements in performance, features, and connectivity options.

At its core, the ESP32 is powered by a dual-core Tensilica Xtensa LX6 microprocessor, which can operate at up to 240 MHz, providing excellent processing capabilities for real-time applications. It comes with 520 KB of SRAM and can support external flash memory, typically ranging from 4 MB to 16 MB.

One of the most significant advantages of the ESP32 is its rich set of connectivity features. It includes integrated Wi-Fi (802.11 b/g/n) and Bluetooth (Classic and BLE), making it ideal for wireless communication without the need for additional modules. This allows developers to build smart devices that can communicate with the internet or with other Bluetooth-enabled devices.

The ESP32 features a wide range of digital and analog interfaces, including GPIOs, I2C, SPI, UART, ADC, DAC, PWM, and touch sensors, offering versatile options for sensor and actuator interfacing. It also includes features like Hall effect sensors, temperature sensors, and secure boot/encryption, adding to its robustness.

It is supported by popular development environments such as the Arduino IDE, ESP-IDF (Espressif IoT Development Framework), and MicroPython, making it accessible to both hobbyists and professional developers.

Before we delve into the actual reasons for the popularity of ESP32, let's take a look at some of its important specifications. The specs listed below belong to the ESP32variant.

- Integrated Crystal– 40 MHz
- Module Interfaces– UART, SPI, I2C, PWM, ADC, DAC, GPIO, pulse counter, capacitive touch sensor
- Integrated SPI flash– 4 MB
- ROM– 448 KB (for booting and core functions)
- SRAM– 520 KB
- Integrated Connectivity Protocols– WiFi, Bluetooth, BLE
- On–chip sensor– Hall sensor
- Operating temperature range– –40 – 85 degrees Celsius
- Operating Voltage– 3.3V
- Operating Current– 80 mA (average)
- ESP32-D0WDQ6 (and ESP32D0WD)
- ESP32-D2WD
- ESP32-S0WD

- System in package (SiP) – ESP32-PICO-D4
- ESP32 S series
- ESP32-C series
- ESP32-H series

The ESP32 is most commonly used for mobile devices, wearable tech, and IoT applications, such as Nabto Edge. Moreover, since Mongoose OS introduced an ESP32 IoT Starter Kit, the ESP32 has gained a reputation as the ultimate chip for hobbyists and IoT developers. It's suitable for commercial IoT, and its capabilities and resources have grown impressively over the past four years.

5.4 Regulated Power Supply:

5.4.1 Introduction:

Power supply is a supply of electrical power. A device or system that supplies electrical or other types of energy to an output load or group of loads is called a power supply unit or PSU. The term is most commonly applied to electrical energy supplies, less often to mechanical ones, and rarely to others.

A power supply may include a power distribution system as well as primary or secondary sources of energy such as

- Conversion of one form of electrical power to another desired form and voltage, typically involving converting AC line voltage to a well-regulated lower-voltage DC for electronic devices. Low voltage, low power DC power supply units are commonly integrated with the devices they supply, such as computers and household electronics.
- Batteries.
- Chemical fuel cells and other forms of energy storage systems.
- Solar power.
- Generators or alternators.

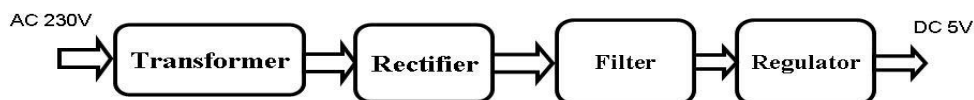


Fig 5.4.1.1 Regulated Power Supply

The basic circuit diagram of a regulated power supply (DC O/P) with led connected as load is shown in fig 5.4.1.1

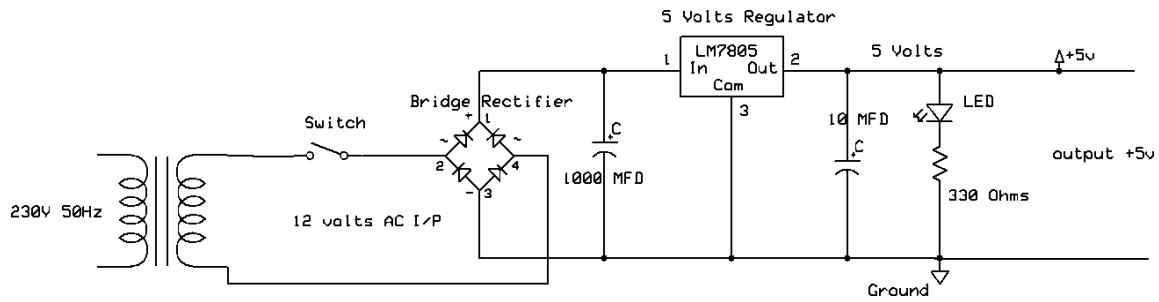


Fig 5.4.1.2 Circuit diagram of Regulated Power Supply with Led connection

The components mainly used in above figure are

- 230V AC MAINS
- TRANSFORMER
- BRIDGE RECTIFIER(DIODES)
- CAPACITOR
- VOLTAGE REGULATOR(IC 7805)
- RESISTOR
- LED(LIGHT EMITTING DIODE)

The detailed explanation of each and every component mentioned above is as follows:

5.5 Buzzer



Fig.5.5.1 buzzer

Basically, the sound source of a piezoelectric sound component is a piezoelectric diaphragm. A piezoelectric diaphragm consists of a piezoelectric ceramic plate which has electrodes on both sides and a metal plate (brass or stainless steel, etc.). A piezoelectric ceramic plate is attached to a metal plate with adhesives. Applying D.C. voltage between electrodes of a piezoelectric diaphragm causes mechanical distortion due to the piezoelectric effect. For a misshaped piezoelectric element, the distortion of the piezoelectric element expands in a radial

direction. And the piezoelectric diaphragm bends toward the direction. The metal plate bonded to the piezoelectric element does not expand. Conversely, when the piezoelectric element shrinks, the piezoelectric diaphragm bends in the direction. Thus, when AC voltage is applied across electrodes, the bending is repeated, producing sound waves in the air.

To interface a buzzer the standard transistor interfacing circuit is used. Note that if a different power supply is used for the buzzer, the 0V rails of each power supply must be connected to provide a common reference.

If a battery is used as the power supply, it is worth remembering that piezo sounders draw much less current than buzzers. Buzzers also just have one 'tone', whereas a piezo sounder is able to create sounds of many different tones.

To switch on buzzer -high 1

To switch off buzzer -low 1

5.6 Voltage Regulator:

A voltage regulator (also called a 'regulator') with only three terminals appears to be a simple device, but it is in fact a very complex integrated circuit. It converts a varying input voltage into a constant 'regulated' output voltage. Voltage Regulators are available in a variety of outputs like 5V, 6V, 9V, 12V and 15V. The LM78XX series of voltage regulators are designed for positive input. For applications requiring negative input, the LM79XX series is used. Using a pair of 'voltage-divider' resistors can increase the output voltage of a regulator circuit.

It is not possible to obtain a voltage lower than the stated rating. You cannot use a 12V regulator to make a 5V power supply. Voltage regulators are very robust. These can withstand over-current draw due to short circuits and also over-heating. In both cases, the regulator will cut off before any damage occurs. The only way to destroy a regulator is to apply reverse voltage to its input. Reverse polarity destroys the regulator almost instantly. Fig: 3.3.11 shows voltage regulator.



fig 5.6.1 voltage regulator

5.7 Vibration Sensor

Vibration sensor used to detect the earth quake alerts in the coal mine. here we are mercury switch acts as vibrating element it can detect any vibrations and automatically alerts the buzzer and data displays into lcd and iot module at same time.



Fig.5.7.1 Vibration sensor

5.8 DHT11- HUMIDITY AND TEMPERATURE SENSOR

DHT11 digital temperature and humidity sensor is a calibrated digital signal output of the temperature and humidity combined sensor. It uses a dedicated digital modules capture technology and the temperature and humidity sensor technology to ensure that products with high reliability and excellent long-term stability. Sensor includes a resistive element and a sense of wet NTC temperature measurement devices, and with a high-performance 8-bit microcontroller connected .

Humidity Sensor is one of the most important devices that has been widely in consumer, industrial, biomedical, and environmental etc. applications for measuring and monitoring Humidity. Humidity is defined as the amount of water present in the surrounding air. This water content in the air is a key factor in the wellness of mankind. For example, we will feel comfortable even if the temperature is 00C with less humidity i.e. the air is dry.

But if the temperature is 100C and the humidity is high i.e. the water content of air is high, then we will feel quite uncomfortable. Humidity is also a major factor for operating sensitive equipment like electronics, industrial equipment, electrostatic sensitive devices and high voltage devices etc. Such sensitive equipment must be operated in a humidity environment that is suitable for the device.

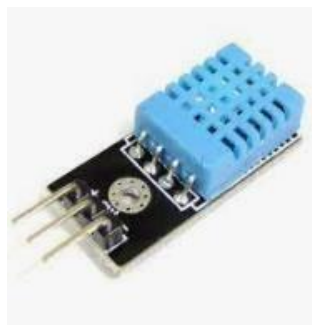


Fig.5.8.1 temperature sensor

Hence, sensing, measuring, monitoring and controlling humidity is a very important task. Some of the important areas of application for sensing, measuring and controlling Humidity are mentioned below.

Domestic: Sensing and controlling humidity in our homes and offices is important as higher humidity conditions will affect the blood flow. Other areas include cooking, indoor plantation etc.

Industrial: In industries like refineries, chemical, metal, or other industries where furnaces are used, high humidity will reduce the amount of oxygen in the air and hence reduces the firing rate. Other industries like food processing, textile, paper etc. also need control of humidity.

Moisture: Generally, the term Moisture means water content of any material or substance. But practically, the term Moisture refers to the water content in solids and liquids. The term Humidity refers to the water content in gases (air).

Absolute Humidity: Absolute Humidity (AH) is the ratio of mass of the water vapour to the volume of the air. If m is the mass of the water vapour and V is the total volume i.e. volume of air and water vapour mixture, then Absolute Humidity AH is given by

$$AH = m/V$$

Absolute Humidity doesn't take temperature in to account but it changes with temperature and pressure.

Relative Humidity: Whenever we talk about measuring Humidity, it usually Relative Humidity that we are talking about (unless otherwise specified).

Relative Humidity or RH is the ratio of the actual water vapour pressure present in the air at a temperature to the maximum water vapour pressure present in the air at the same temperature.

In weather reports and forecasts, the probability of precipitation or dew or fog is indicated using Relative Humidity and hence, it is considered an important metric.

Relative Humidity takes both temperature and pressure in to consideration. Hence, the Humidity Sensors which measure Relative Humidity, measure both the moisture content as well as the air temperature.

NOTE: For temperatures above 1000C, measuring Relative Humidity (RH) is of no use as it would deliver misleading values.

Specific Humidity: Specific Humidity (SH) is the ratio of mass of the water vapor to the total mass of the air.

Mixing Ratio or Humidity Ratio: Mixing Ratio is the ratio of mass of the water vapor to mass of the dry air.

Dew Point Temperature: Dew Point Temperature is the temperature at which the water vapor content is saturated in the air. At Dew Point temperature, the Relative Humidity RH is 100%. In other words, for the air to hold maximum amount of water vapor (or moisture), it has to reach Dew Point Temperature.

5.8.1.2 DHT11

We have two versions of the DHT sensor, they look a bit similar and have the same pinout, but have different characteristics. Here are the specs:

- Ultra low cost

- 3 to 5V power and I/O
- 2.5mA max current use during conversion (while requesting data)
- Good for 20-80% humidity readings with 5% accuracy
- Good for 0-50°C temperature readings $\pm 2^{\circ}\text{C}$ accuracy
- No more than 1 Hz sampling rate (once every second)
- Body size 15.5mm x 12mm x 5.5mm
- 4 pins with 0.1" spacing

1. Working and storage conditions The proposed scope of work may result in up to 3% RH temporary drift of the signal. Return to normal working conditions, the sensor calibration status will slowly recover. To speed up the recovery process can be found in "recovery process." The use of the product will accelerate the aging process for a long time under abnormal operating conditions. Avoid placing components on a long-term condensation and dry conditions and the following environments. A. smoke B. Acid or oxidizing gases such as sulfur dioxide, hydrochloric acid Recommended Storage Environment Temperature : 10~40°C Humidity : 60% RH or less

2. Effects of exposure to chemical substances Sensing resistive humidity sensor will be disturbed chemical vapor layer, the diffusion layer in the induction of chemicals may cause drift and measurement sensitivity. In a clean environment, slowly release contaminants out. The recovery process described below to accelerate the process.

3. Temperature Effect Relative humidity of the gas is largely dependent on temperature. Therefore, when measuring the humidity should be possible to ensure that the humidity sensor works at the same temperature. If you share a printed circuit board with electronic components heat released in the sensor should be installed as far as possible away from the electronic components, and installed at the bottom of the heat source, while maintaining a well-ventilated enclosure. To reduce the thermal conductivity sensor and a copper plating layer of the printed circuit board should be as minimal other portions, and leaving a gap between them.

4. Light effects Prolonged exposure to sunlight or strong ultraviolet radiation, will reduce performance.

5. Recovery process Placed under extreme operating conditions or chemical vapor sensors, through the following process, you can return it to the state calibration. 45 °C and humidity under 70% RH conditions were maintained for more than 5 hours.

6. Wiring Precautions DATA signal wire quality will affect the communication distance and communication quality, we recommend using a high-quality shielded cable.

7. Soldering Information Manual welding, at a temperature of 300 °C maximum contact time must be less than 10 seconds. 8. Product upgrades For details, please consult our technical department.

5.9 CO SENSOR:

A carbon monoxide detector or CO detector is a device that detects the presence of the carbon monoxide (CO) gas in order to prevent carbon monoxide poisoning. In the late 1990s Underwriters Laboratories (UL) changed their definition of a single station CO detector with a sound device in it to a carbon monoxide (CO) alarm. This applies to all CO safety alarms that meet UL 2034; however for passive indicators and system devices that meet UL 2075, UL refers to these as carbon monoxide detectors. CO is a colorless, tasteless and odorless compound produced by incomplete combustion of carbon-containing materials. It is often referred to as the "silent killer" because it is virtually undetectable without using detection technology and, in a study by Underwriters Laboratories, "Sixty percent of Americans could not identify any potential signs of a CO leak in the home".^[1] Elevated levels of CO can be dangerous to humans depending on the amount present and length of exposure. Smaller concentrations can be harmful over longer periods of time while increasing concentrations require diminishing exposure times to be harmful.

CO detectors are designed to measure CO levels over time and sound an alarm before dangerous levels of CO accumulate in an environment, giving people adequate warning to safely ventilate the area or evacuate. Some system-connected detectors also alert a monitoring service that can dispatch emergency services if necessary.

While CO detectors do not serve as smoke detectors and vice versa, dual smoke/CO detectors are also sold. Smoke detectors detect the smoke generated by flaming or smoldering fires, whereas CO detectors detect and warn people about dangerous CO buildup caused, for example, by a malfunctioning fuel-burning device. In the home, some common sources of CO include open flames, space heaters, water heaters, blocked chimneys or running a car inside a garage.

5.10 LED:

A light-emitting diode (LED) is a semiconductor light source. LED's are used as indicator lamps in many devices, and are increasingly used for lighting. Introduced as a practical electronic component in 1962, early LED's emitted low-intensity red light, but modern versions are available across the visible, ultraviolet and infrared wavelengths, with very high brightness. The internal structure and parts of a led are shown below.

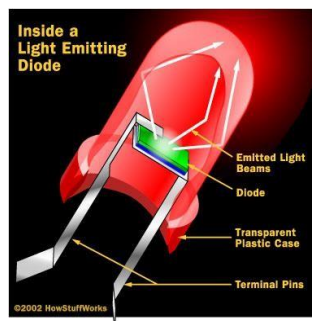


Fig.5.10.1 Inside a LED

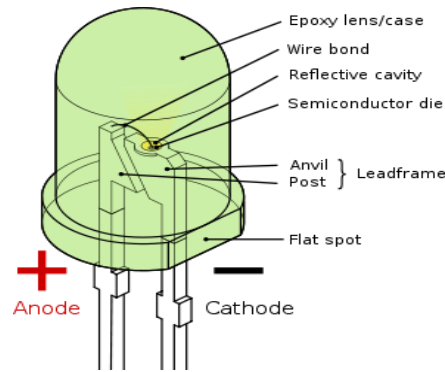


Fig 5.10.2 Parts of a LED

5.10.2.1 WORKING OF LED

The structure of the LED light is completely different than that of the light bulb. Amazingly, the LED has a simple and strong structure. The light-emitting semiconductor material is what determines the LED's color. The LED is based on the semiconductor diode.

When a diode is forward biased (switched on), electrons are able to recombine with holes within the device, releasing energy in the form of photons. This effect is called electroluminescence and the color of the light (corresponding to the energy of the photon) is determined by the energy gap of the semiconductor. An LED is usually small in area (less than 1 mm^2), and integrated optical components are used to shape its radiation pattern and assist in reflection. LED's present many advantages over incandescent light sources including lower energy consumption, longer lifetime, improved robustness, smaller size, faster switching, and greater durability and reliability. However, they are relatively expensive and require more precise current and heat management than traditional light sources. Current LED products for general lighting are more expensive to buy than fluorescent lamp sources of comparable output. They also enjoy use in applications as diverse as replacements for traditional light sources in automotive lighting (particularly indicators) and in traffic signals. The compact size of LED's has allowed new text and video displays and sensors to be developed, while their high switching rates are useful in advanced communications technology.

5.11 ZIGBEE MODULE



Fig.5.11.1 zigbee module

The **Zigbee module** is a wireless communication device based on the **IEEE 802.15.4 standard**, designed for low-power, low-data-rate, and short-range communication. It is widely used in applications such as **home automation, industrial monitoring, smart energy**

systems, and Internet of Things (IoT) due to its reliability, low energy consumption, and mesh networking capability.

A Zigbee module typically consists of a **microcontroller**, a **radio transceiver**, and a **firmware stack** that handles the Zigbee protocol. Some common Zigbee module manufacturers include **Digi International (XBee series)**, **Texas Instruments**, and **Silicon Labs**. These modules are available in various configurations, including surface mount and plug-and-play versions, making integration easy for different hardware designs.

One of Zigbee's key strengths is its support for **mesh networking**, where each node can communicate with its neighbors to route data efficiently. This ensures increased range, reliability, and redundancy, as data can hop through multiple nodes to reach its destination. The network structure typically consists of three types of devices:

1. **Coordinator** – Initializes the network, assigns addresses, and manages routing.
2. **Router** – Forwards data and can also serve as an intermediate node.
3. **End Device** – Low-power device that communicates with the router or coordinator but does not forward data.

Zigbee operates in the **2.4 GHz ISM band** globally, and also in the **868 MHz (Europe)** and **915 MHz (USA)** bands for specific applications. It supports a data rate of up to **250 kbps**, which is sufficient for sensor readings, control signals, and status updates in most embedded systems.

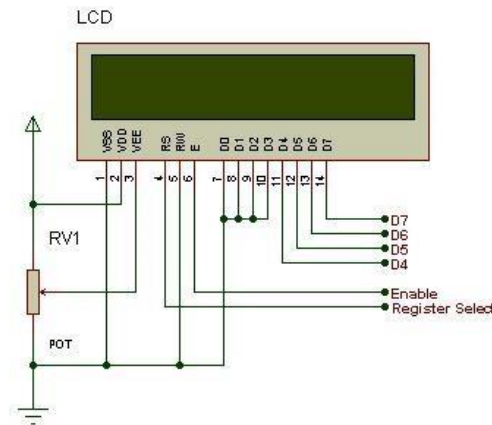
Zigbee modules are designed for **ultra-low power consumption**, making them ideal for battery-operated devices. They support various sleep modes and can wake up quickly to transmit or receive data, thus conserving energy.

The Zigbee protocol also includes built-in **security features** such as **128-bit AES encryption**, **network authentication**, and **device authorization**, ensuring safe communication in sensitive environments.

Zigbee modules can be easily configured using **AT commands** or programmable firmware. Development tools and libraries are available for various microcontrollers and platforms, enabling fast prototyping and deployment.

In conclusion, the Zigbee module is a robust and energy-efficient solution for wireless communication in applications that demand secure, reliable, and scalable networks. Its mesh networking capability, low power usage, and flexible architecture make it a popular choice for modern IoT and automation systems.

5.12 LCD DISPLAY



5.12.1 pin daigram of LCD

The below table 5.12.1.1 shows LCD (Pin description)

Pin No	Pin Name	Description
1	VSS	Ground (0V)
2	VCC	Power supply (+5V)
3	VEE	Contrast adjustment (via potentiometer)
4	RS	Register Select: 0 = Instruction, 1 = Data
5	R/W	Read/Write: 0 = Write, 1 = Read
6	EN	Enable signal to latch data/command
7	D0	Data bus line 0 (Least Significant Bit)
8	D1	Data bus line 1
9	D2	Data bus line 2
10	D3	Data bus line 3
11	D4	Data bus line 4
12	D5	Data bus line 5
13	D6	Data bus line 6
14	D7	Data bus line 7

Fig 5.12.1.1 pin description

This project uses a character LCD to display real-time water level and drainage status information. The LCD requires 3 control lines plus either 4 or 8 data lines depending on whether we choose 4-bit or 8-bit mode. In this project, we opt for the 4-bit mode to reduce the number of GPIO pins used on the ESP32.

Control Line Details:

- EN (Enable):

The microcontroller pulls EN low, sets RS and data lines, then pulls EN high briefly to latch the data/command into the LCD. After a short delay, EN is pulled low again to execute the instruction. This pulse signals the LCD to process the data on the bus.

- RS (Register Select):

- RS = 0: Instruction/command (e.g., clear screen, cursor position).
- RS = 1: Data to be displayed (characters like numbers or letters).
- RW (Read/Write):

Tied permanently to 0 (write mode) in our project. This simplifies wiring and avoids bus conflicts but disables reading the LCD's busy flag. Instead, we use fixed delays after commands to allow the LCD time to process instructions.

Data Bus:

- Using 4-bit mode, the LCD data pins DB4 to DB7 connect to 4 GPIO pins on the ESP32.
- Data is sent in two 4-bit "nibbles" for each byte (high nibble first, then low nibble).

LCD Pin Requirements:

- Control Lines (3):
- EN (Enable): Tells the LCD to read the data on the data bus.
- RS (Register Select): Selects whether the data is a command (0) or display character (1).
- RW (Read/Write): Selects between writing data to the LCD (0) or reading from it (1). In our project, RW is hardwired to write mode (0) to simplify connections.
- Data Lines (4 or 8):

For 4-bit mode, we use 4 data lines (DB4 to DB7). This minimizes ESP32 pin usage and still provides efficient communication.

CHAPTER 6

SIMULATION TOOLS

This project is implemented using following software's:

- Express PCB – for designing circuit
- Arduino IDE compiler - for compilation part
- Proteus 7 (Embedded C) – for simulation part

6.1 EXPRESS PCB:

Breadboards are great for prototyping equipment as it allows great flexibility to modify a design when needed; however the final product of a project, ideally should have a neat PCB, few cables, and survive a shake test. Not only is a proper PCB neater but it is also more durable as there are no cables which can yank loose.

Express PCB is a software tool to design PCBs specifically for manufacture by the company Express PCB (no other PCB maker accepts Express PCB files). It is very easy to use, but it does have several limitations.

- It can be likened to more of a toy then a professional CAD program.
- It has a poor part library (which we can work around)
- It cannot import or export files in different formats
- It cannot be used to make prepare boards for DIY production

Express PCB has been used to design many PCBs (some layered and with surface-mount parts. Print out PCB patterns and use the toner transfer method with an Etch Resistant Pen to make boards. However, Express PCB does not have a nice print layout. Here is the procedure to design in Express PCB and clean up the patterns so they print nicely.

6.1.1 Preparing Express PCB for First Use:

Express PCB comes with a less then exciting list of parts. So before any project is started head over to Audio logical and grab the additional parts by morsel, ppl, and tangent, and extract them into your Express PCB directory. At this point start the program and get ready to setup the workspace to suit your style.

Click View -> Options. In this menu, setup the units for “mm” or “in” depending on how you think, and click “see through the top copper layer” at the bottom. The standard color scheme of red and green is generally used but it is not as pleasing as red and blue.

6.1.2 The Interface:

When a project is first started you will be greeted with a yellow outline. This yellow outline is the dimension of the PCB. Typically after positioning of parts and traces, move them to their final position and then crop the PCB to the correct size. However, in designing a board with a certain size constraint, crop the PCB to the correct size before starting.

Fig: 6.1.2.1 show the toolbar in which the each button has the following functions:



Fig 6.1.2.1 Tool bar necessary for the interface

- The select tool: It is fairly obvious what this does. It allows you to move and manipulate parts. When this tool is selected the top toolbar will show buttons to move traces to the top / bottom copper layer, and rotate buttons.
- The zoom to selection tool: does just that.
- The place pad: button allows you to place small solder pads which are useful for board connections or if a part is not in the part library but the part dimensions are available. When this tool is selected the top toolbar will give you a large selection of round holes, square holes and surface mount pads.
- The place component: tool allows you to select a component from the top toolbar and then by clicking in the workspace places that component in the orientation chosen using the buttons next to the component list. The components can always be rotated afterwards with the select tool if the orientation is wrong.
- The place trace: tool allows you to place a solid trace on the board of varying thicknesses. The top toolbar allows you to select the top or bottom layer to place the trace on.
- The Insert Corner in trace: button does exactly what it says. When this tool is selected, clicking on a trace will insert a corner which can be moved to route around components and other traces.
- The remove a trace button is not very important since the delete key will achieve the same result.

6.2 Design Considerations:

Before starting a project there are several ways to design a PCB and one must be chosen to suit the project's needs.

Single sided, or double sided?

When making a PCB you have the option of making a single sided board, or a double sided board. Single sided boards are cheaper to produce and easier to etch, but much harder to design for large projects. If a lot of parts are being used in a small space it may be difficult to make a single sided board without jumpering over traces with a cable. While there's technically nothing wrong with this, it should be avoided if the signal travelling over the traces is sensitive (e.g. audio signals).

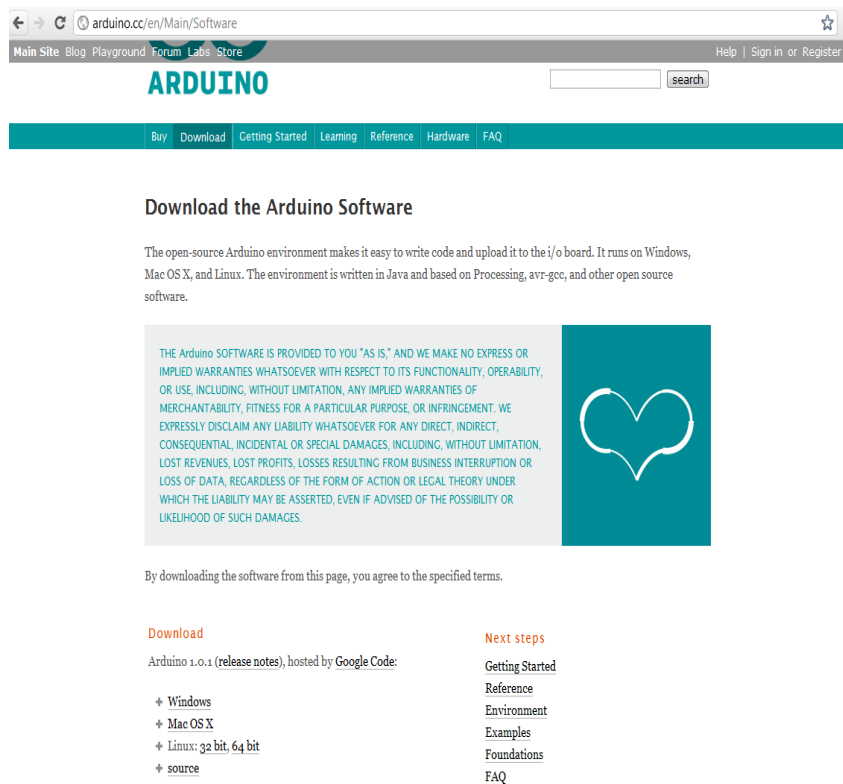
A double sided board is more expensive to produce professionally, more difficult to etch on a DIY board, but makes the layout of components a lot smaller and easier. It should be noted that if a trace is running on the top layer, check with the components to make sure you can get to its pins with a soldering iron. Large capacitors, relays, and similar parts which don't have axial leads can NOT have traces on top unless boards are plated professionally.

Ground-plane or other special purposes for one side When using a double sided board you must consider which traces should be on what side of the board. Generally, put power traces on the top of the board, jumping only to the bottom if a part cannot be soldered onto the top plane (like a relay), and vice- versa.

Some projects like power supplies or amps can benefit from having a solid plane to use for ground. In power supplies this can reduce noise, and in amps it minimizes the distance between parts and their ground connections, and keeps the ground signal as simple as possible. However, care must be taken with stubborn chips such as the TPA6120 amplifier from TI. The TPA6120 datasheet specifies not to run a ground plane under the pins or signal traces of this chip as the capacitance generated could effect performance negatively.

6.3 ARDUINO COMPILING

STEP1:



The screenshot shows the Arduino.cc website's software download page. The browser address bar displays 'arduino.cc/en/Main/Software'. The site's navigation bar includes links for 'Main Site', 'Blog', 'Playground', 'Forum', 'Labs', and 'Store', along with a search bar and links for 'Help', 'Sign in', and 'Register'. A teal header bar contains links for 'Buy', 'Download', 'Getting Started', 'Learning', 'Reference', 'Hardware', and 'FAQ'. The main heading is 'Download the Arduino Software'. Below it, a paragraph states: 'The open-source Arduino environment makes it easy to write code and upload it to the i/o board. It runs on Windows, Mac OS X, and Linux. The environment is written in Java and based on Processing, avr-gcc, and other open source software.' A light gray box contains a disclaimer: 'THE Arduino SOFTWARE IS PROVIDED TO YOU "AS IS," AND WE MAKE NO EXPRESS OR IMPLIED WARRANTIES WHATSOEVER WITH RESPECT TO ITS FUNCTIONALITY, OPERABILITY, OR USE, INCLUDING, WITHOUT LIMITATION, ANY IMPLIED WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, OR INFRINGEMENT. WE EXPRESSLY DISCLAIM ANY LIABILITY WHATSOEVER FOR ANY DIRECT, INDIRECT, CONSEQUENTIAL, INCIDENTAL OR SPECIAL DAMAGES, INCLUDING, WITHOUT LIMITATION, LOST REVENUES, LOST PROFITS, LOSSES RESULTING FROM BUSINESS INTERRUPTION OR LOSS OF DATA, REGARDLESS OF THE FORM OF ACTION OR LEGAL THEORY UNDER WHICH THE LIABILITY MAY BE ASSERTED, EVEN IF ADVISED OF THE POSSIBILITY OR LIKELIHOOD OF SUCH DAMAGES.' To the right of this text is a teal square with a white heart icon. Below the disclaimer, a line reads: 'By downloading the software from this page, you agree to the specified terms.' Two columns of links are provided: 'Download' (with links for Windows, Mac OS X, Linux: 32 bit, 64 bit, and source) and 'Next steps' (with links for Getting Started, Reference, Environment, Examples, Foundations, and FAQ).

Download the Arduino Software

The open-source Arduino environment makes it easy to write code and upload it to the i/o board. It runs on Windows, Mac OS X, and Linux. The environment is written in Java and based on Processing, avr-gcc, and other open source software.

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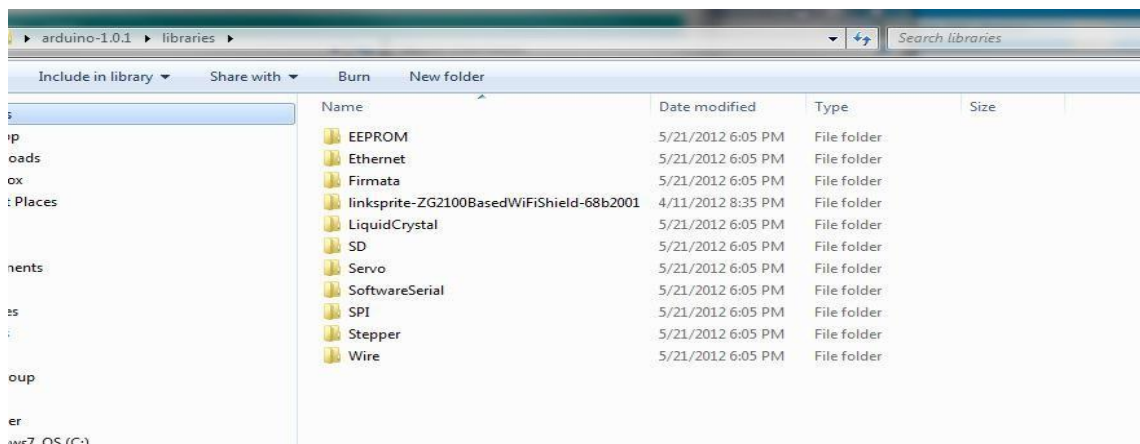
Arduino 1.0.1 ([release notes](#)), hosted by [Google Code](#):

- ✦ [Windows](#)
- ✦ [Mac OS X](#)
- ✦ [Linux: 32 bit, 64 bit](#)
- ✦ [source](#)

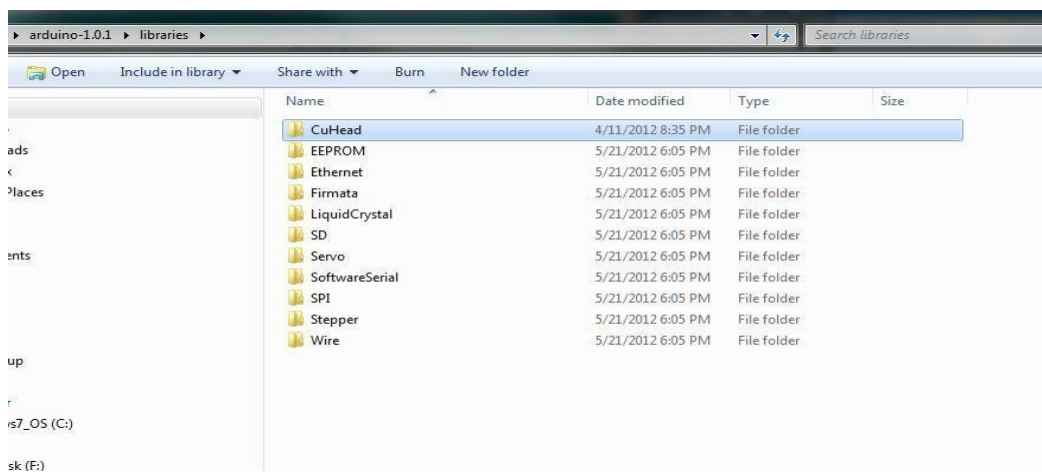
Next steps

- [Getting Started](#)
- [Reference](#)
- [Environment](#)
- [Examples](#)
- [Foundations](#)
- [FAQ](#)

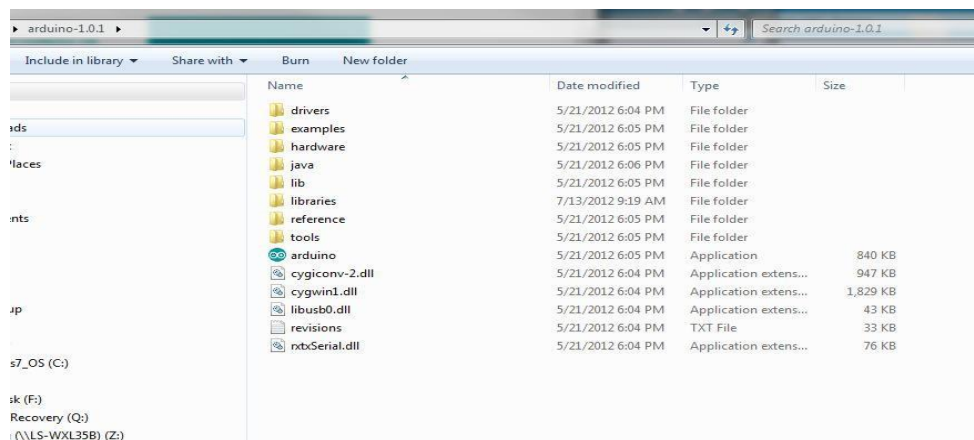
STEP2:



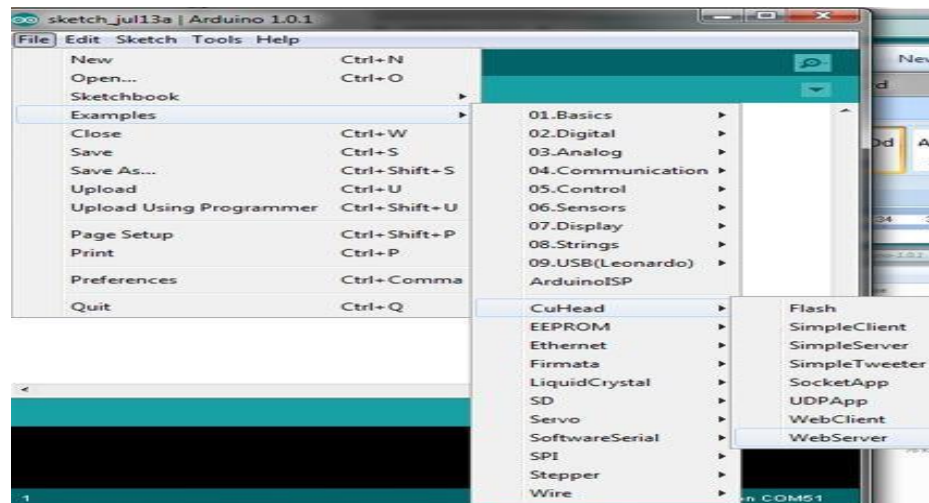
STEP3:



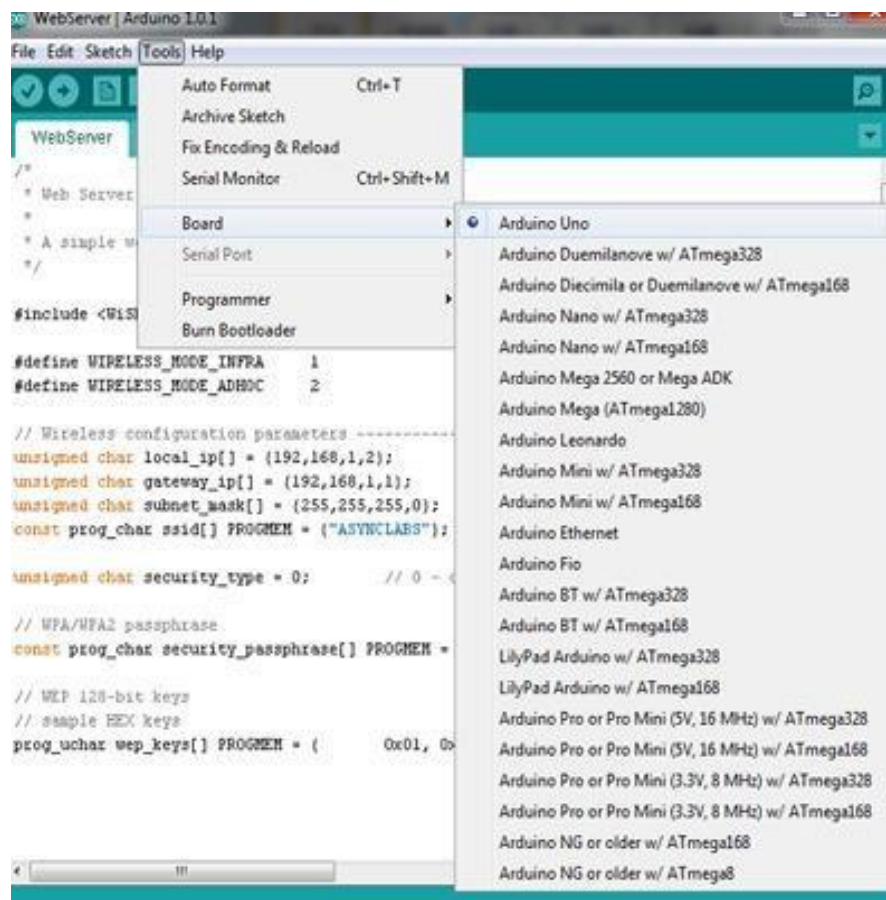
STEP 4: Launch Arduino by double click “arduino” below



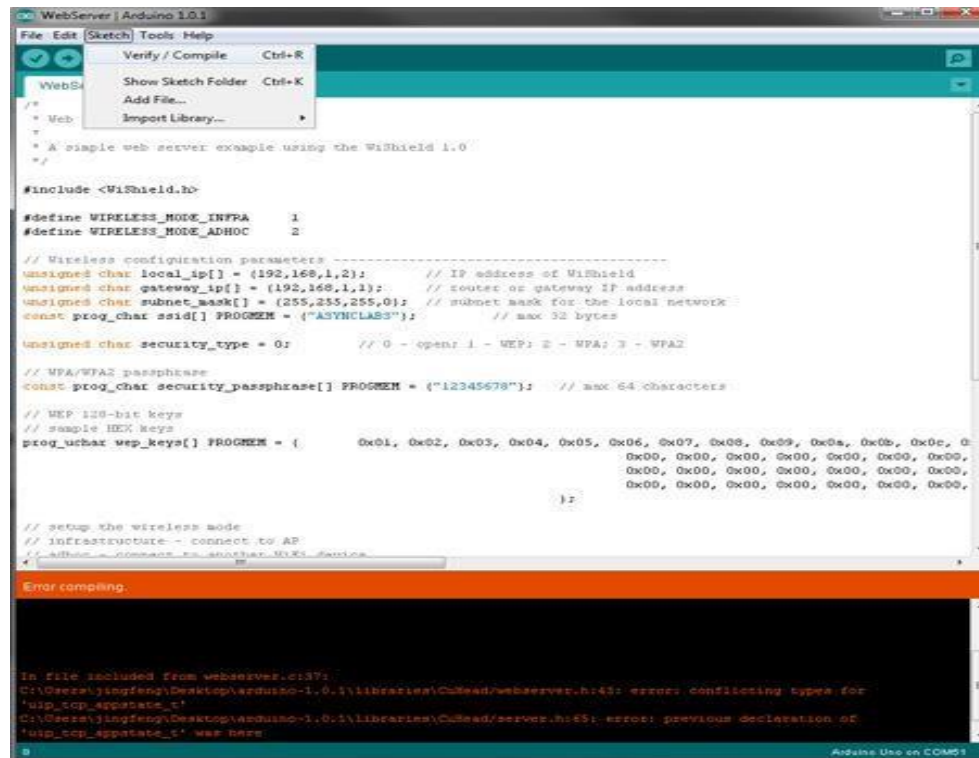
6.3.1. EXAMPLE



STEP1: Select the target board as “Arduino Uno”:



STEP 2: Click Sketch-> Verify/Compile:



CHAPTER 7

RESULTS

This section presents the results of the proposed system

7.1 Output of Transmitter section

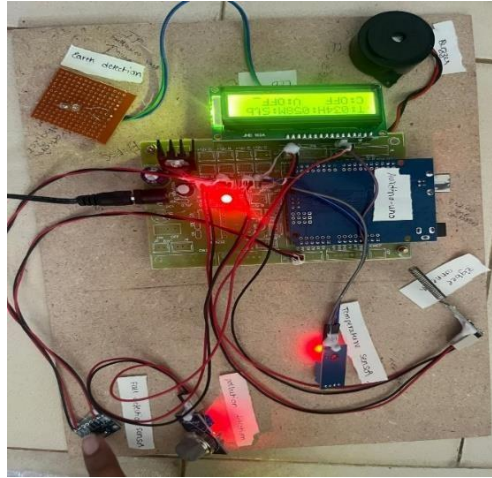


Fig.7.1.1 Transmitter section

The fig 7.1.1 presents a functional prototype of an IoT-enabled safety monitoring system, mounted on a simple wooden board. At its heart sits a microcontroller development board—most likely an Arduino—responsible for polling multiple environmental and motion sensors, driving an LCD panel, and triggering an audible alarm.

At the top of the board, a green backlit 16×2 character LCD displays live readings: in the example shown, it reports “Temp: 30 °C,” “GAS: 852,” with status flags for carbon monoxide (“CO: OFF”) and vibration (“VIB: OFF”). To the right of the LCD, a small piezo buzzer provides an audible alert if any measured parameter crosses a predefined safety threshold.

Surrounding the central controller are five distinct sensors, each connected via jumper wires and labeled with handwritten tags:

1. **Gas Sensor** (bottom-right): Detects concentrations of hazardous gases such as methane or carbon monoxide, critical for early warning in confined or subterranean environments.
2. **Temperature Sensor** (just above the gas sensor): Monitors ambient heat; sudden rises could indicate a fire hazard or equipment overheating.
3. **Vibration Sensor** (center-right): Senses mechanical vibrations or shocks, potentially signaling cave-ins, heavy machinery incidents, or fall impacts.
4. **Fall/Tilt Sensor** (bottom-left): Likely an accelerometer or tilt switch mounted on a small breakout board; it flags if the helmet wearer has fallen or remains motionless in an unusual orientation.
5. **Arm/Presence Detection** (top-left): A simple contact switch or infrared proximity module that confirms the device is properly worn, ensuring the system only activates when on a user’s helmet.

7.1.1.1 Output of LCD Display Transmitter section



fig 7.1.1.2 transmitter output

The fig 7.1.12 shows A close-up of a green backlit 16×2 character LCD module wired into a microcontroller-based safety prototype. The top line reads sensor data—“T:034” (34 °C), “H:058” (58 % humidity), and “M:Stb” (motion status: stable)—while the bottom line shows two binary alerts: “C:OFF” (carbon monoxide alarm off) and “V:OFF” (vibration alarm off). The display’s pins and mounting holes are visible along its top flange. This panel provides real-time environmental and motion feedback, forming the human-machine interface of an IoT-enabled safety monitoring system for hazardous work environments.

7.2 Output of Reciver section

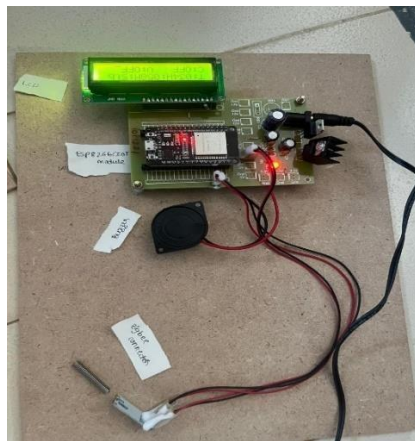


Fig.7.2.1 reciever section

The fig 7.2.1 shows a near-complete IoT safety-monitoring prototype mounted on a simple wooden base. Atop the board sits a green-backlit 16×2 LCD module, wired into the main controller to display real-time readings. Just below it, an **ESP8266 NodeMCU** development board (labeled “ESP8266-NodeMCU”) serves as the wireless brains of the unit, gathering sensor data and sending it over Wi-Fi to a remote server or smartphone app.

To the right of the NodeMCU is the power circuitry: a barrel-jack jack connector, a 5 V regulator, filter capacitors, and indicator LEDs ensure the system runs stably from a

standard DC adapter. In front of the module, a black **piezo buzzer** (labeled “Buzzer”) provides audible alarms if any monitored threshold is breached.

Two wired sensors hang off the bottom edge:

- A **current-clamp sensor** (labeled “Engine current sensor”) for measuring electrical draw—useful for detecting stalled or overloaded machinery.
- A **vibration or tilt sensor** packaged in a small white housing, which could detect falls, abnormal equipment movement, or dislodged cables.

All sensors connect via jumper wires into the NodeMCU’s analog and digital pins. The firmware continuously reads temperature, gas concentration, vibration, and current data, updating the LCD and sounding the buzzer if dangerous conditions are detected. By leveraging the ESP8266’s Wi-Fi, the system can also push alerts to mine supervisors, construction managers, or maintenance teams in real time—enhancing worker safety in hazardous environments.

7.2.1.1 Output of LCD Display Receiver section



fig.7.2.1.2 receiver output

The fig 7.2.1.2A close-up of a green-backlit 16×2 character LCD integrated into an IoT safety prototype. The top line reads “T:034” (34 °C), “H:058” (58 % humidity), and “M:Fall” (motion status: fall detected). The bottom line shows “C:ON” (carbon monoxide alarm active) and “V:OFF” (vibration alarm inactive). White silicone standoffs secure the display to the underlying PCB, and soldered header pins and ribbon cables connect it to a microcontroller. This interface provides instantaneous environmental and motion alerts—fall detection and gas hazard—allowing workers or supervisors to respond swiftly to dangerous conditions.

7.3 IoT sever vibration sensor output

S.No	Temperature	Humidity	Co	Vib	MEMS_Status	Date
1	34	52	OFF	OFF	Stable	2025-04-22 11:56:04
2	34	53	OFF	OFF	Stable	2025-04-22 11:55:44
3	34	55	OFF	ON	Stable	2025-04-22 11:55:24
4	34	60	OFF	OFF	Stable	2025-04-22 11:55:03
5	34	68	OFF	OFF	Stable	2025-04-22 11:54:43
6	35	81	OFF	OFF	Stable	2025-04-22 11:54:23

fig.7.3.1 vibration sensor output

The table 7.3.1 readings captured on April 22, 2025.

Columns include:

- S.No: Record index
- Temperature (°C)
- Humidity (%)
- CO: Carbon monoxide alarm status
- Vib: Vibration alarm status
- MEMS_Status: Accelerometer-based motion state
- Date: Timestamp

At 11:56:04 the system reported 34 °C and 52 % humidity with both CO and vibration alarms off and a “Stable” motion state. Subsequent entries show temperatures between 34–35 °C and humidity rising from 53 % to 81 %, with CO always off. Only the third row triggers a vibration alert (“ON”) while still marked “Stable.” This dataset provides real-time environmental monitoring and mechanical disturbance detection.

7.4 IoT sever vibration sensor output

S.No	Temperature	Humidity	Co	Vib	MEMS_Status	Date
1	35	82	OFF	OFF	Stable	2025-04-22 11:53:42
2	35	77	OFF	OFF	Stable	2025-04-22 11:53:21
3	33	51	OFF	OFF	Stable	2025-04-22 11:52:59
4	33	51	OFF	OFF	Stable	2025-04-22 11:52:39
5	33	51	OFF	OFF	Stable	2025-04-22 11:52:19
6	33	51	OFF	OFF	Stable	2025-04-22 11:51:59

fig.7.4.1 tempature and humidity ouput

Figure 7.4.1 presents a six-entry log of temperature and humidity along side safety flags, timestamped April 22, 2025:

- S.No: Sequential record number
- Temperature: Ranges from 33 °C to 35 °C
- Humidity: Drops from 82 % to 51 % over the six readings
- CO: Carbon monoxide alarm remains OFF throughout
- Vib: Vibration alarm remains OFF throughout
- MEMS_Status: All entries report “Stable,” indicating no fall or excessive movement
- Date: Timestamps span 11:53:42 to 11:51:59

The data show a slight cooling (35 °C → 33 °C) paired with significant dehumidification (82 % → 51 %), while safety alarms stay inactive. The consistent “Stable”

status confirms no detected falls or shocks during this interval, making it a baseline environmental snapshot under normal operating conditions.

7.5 IoT sever vibration sensor output

S.No	Temperature	Humidity	Co	Vib	MEMS_Status	Date
1	33	51	OFF	OFF	Stable	2025-04-22 11:51:38
2	33	51	OFF	OFF	Stable	2025-04-22 11:51:18
3	33	51	OFF	OFF	Stable	2025-04-22 11:50:58
4	33	51	OFF	OFF	Fall	2025-04-22 11:50:36
5	33	51	OFF	OFF	Fall	2025-04-22 11:50:15
6	33	51	OFF	OFF	Stable	2025-04-22 11:49:50
7	33	51	OFF	OFF	Stable	2025-04-22 11:49:30

fig.7.5.1 fall detection sensor output

Figure 7.5.1 is readings from the fall-detection sensor on April 22, 2025:

- Temperature and Humidity remain constant at 33 °C and 51 %.
- CO and Vib alarms are OFF for all entries.
- MEMS_Status (“Stable” or “Fall”) indicates detected motion: entries 4 (11:50:36) and 5 (11:50:15) register “Fall,” while the others remain “Stable.”
- Timestamps span from 11:51:38 down to 11:49:50, with readings every ~20 seconds.

This output shows the device correctly distinguishing two fall events amid otherwise stable conditions, demonstrating reliable real-time accident detection.

7.6 IoT Enabled Coal mine safety helmet

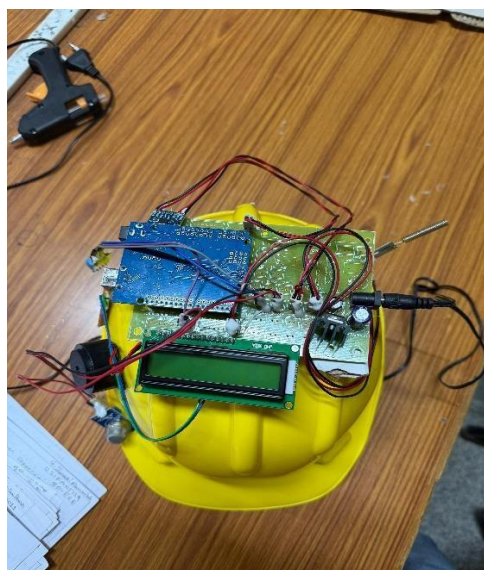


fig 7.6.1 coal mine safety helmet

The figure 7.6.1 showcases the complete IoT safety system mounted on a bright yellow miner's helmet. A custom PCB and microcontroller board—likely an Arduino shield—are secured atop the helmet's crown. A green-backlit 16×2 LCD faces forward to display live readings. Tangle-free standoffs elevate the board, while red and black jumper wires connect peripheral sensors. On the right brim sits a gas sensor module for methane or CO detection. A tilt/vibration sensor assembly and piezo buzzer are also wired in. A voltage-regulated power module feeds the circuitry from a compact battery pack concealed under the helmet shell. This integration delivers real-time environmental and fall alerts directly at the worker's head.

The IoT-enabled coal mine safety helmet enhances miner safety by integrating real-time gas detection, health monitoring, and communication technologies. It detects hazardous gases like methane and carbon monoxide, alerting workers and control rooms instantly. The helmet tracks worker location using GPS or RFID, aiding in quick rescues during emergencies. It monitors vital signs such as heart rate and body temperature, and detects falls or impacts using motion sensors. With wireless communication, it enables constant contact with surface teams. Additionally, it logs data for safety analysis and compliance. Overall, it plays a vital role in ensuring safer and smarter mining operations.

CHAPTER 8

CONCLUSION AND FUTURE SCOPE

8.1 CONCLUSION

In conclusion, the proposed coal mine safety helmet system significantly enhances worker safety through a two-tier architecture combining real-time sensing and wireless data transmission. The transmitter unit, built around an Arduino microcontroller, collects critical data such as vibration, carbon monoxide levels, temperature, humidity, and fall detection using various sensors, and transmits it wirelessly via Zigbee. The receiver unit, powered by the ESP32 microcontroller, receives this data through Zigbee and leverages IoT capabilities to display real-time information on an LCD, activate alarms through a buzzer, and send updates to a remote server for continuous monitoring. This integrated solution provides a robust and intelligent safety mechanism, ensuring timely alerts and effective response to hazardous conditions in coal mining environments.

8.1.1 LIMITATIONS

The IoT-enabled safety helmet, while innovative, faces several challenges. Sensor accuracy can be affected by underground dust, humidity, and temperature extremes. Battery limitations restrict continuous use without frequent recharging. Unreliable network connectivity in deep mines can delay real-time data transmission. High system costs may hinder adoption in smaller mining operations. Added weight from sensors can reduce helmet comfort and usability. Harsh conditions may damage components, requiring frequent maintenance. False alarms due to sensor interference can reduce trust in the system. Lack of standard integration with existing mine safety systems is a barrier. Finally, most systems lack field validation, raising concerns about real-world reliability.

8.2 FUTURE SCOPE

Future research should focus on improving sensor accuracy and durability for harsh mining environments. Developing energy-efficient designs or energy harvesting methods can extend operational life. Integration of AI and edge computing can enable predictive safety alerts and local decision-making. Enhancing network technologies like LoRaWAN or mesh systems can improve underground connectivity. More ergonomic helmet designs are needed to balance safety with comfort. Standardization across systems can enable easier integration with mine management platforms. Research should also explore cost-effective solutions for scalability. Incorporating advanced analytics can help in long-term hazard pattern identification. Improved data encryption is essential for secure communication. Finally, extensive field trials are needed to validate system performance in real mining conditions.

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- [3] Rahim and Tores (2022) - "Augmented Reality Applications for Underground Mining Safety and Productivity" Prototyped AR displays integrated into mining helmets, enabling visual overlay of hazard warnings, equipment maintenance procedures, and emergency escape routes in real-time.
- [4] Karim et al. (2022) - "Methane Ignition Prevention Through Intrinsically Safe Wearable Electronics Design" Detailed design approaches for ensuring electronic components in smart helmets cannot serve as ignition sources in explosive atmospheres, with specific focus on battery and sensor systems.
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- [7] Martinez and Lee (2021) - "Battery Technology Optimization for Wearable Mining Safety Devices" Explored power management strategies for electronic safety systems, achieving 12+ hour operational times through combination of energy harvesting, battery chemistry selection, and adaptive power management.
- [8] Yan et al. (2021) - "Extreme Temperature Performance of Safety Helmet Materials in Deep Mining Applications" Investigated material degradation at depth-associated elevated temperatures, recommending composition modifications to maintain protective capabilities in thermally challenging environments.
- [8] Liu et al. (2020) - "Finite Element Analysis of Impact Energy Absorption in Mining Safety Helmets" Used computational modeling to analyze energy dissipation patterns in various helmet designs, identifying optimal shell thickness and suspension geometries to maximize protection against falling objects common in underground mining environments.
- [9] Chang and DuPont (2020) - "Testing Methodologies for Next-Generation Mining PPE Certification" Proposed updated testing protocols for smart helmet technologies not adequately

addressed by existing standards, including sensor accuracy verification and electronic system resilience.

[10] Nakamura and Singh (2020) - "Self-Healing Polymer Applications in Industrial Safety Equipment" Demonstrated novel materials capable of maintaining structural integrity after minor impacts through autonomic healing mechanisms, with specific application to mining helmet shells.

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[14] Williams et al. (2019) - "Integration Frameworks for Comprehensive Underground Safety Systems" Proposed architectural models for connecting personal protective equipment, environmental monitoring networks, and mine management systems into unified safety ecosystems with helmets as key interface components

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to advanced helmet technologies, including training approaches, technical challenges, and measurable safety improvements.

[19]1Lewis and Garcia (2018) - "Visibility Enhancement Systems for Low-Light Mining Environments"Compared various lighting technologies for mining helmets, establishing optimal specifications for LED systems based on beam pattern, color temperature, and battery efficiency.

[20] Johansson et al. (2017) - "Economic Analysis of Advanced Safety Equipment ImplementationinUndergroundMining"Quantified return-on-investment for various safety helmet technologies, demonstrating that while initial procurement costs were higher, total operational costs decreased through reduced injuries and associated downtime.

APPENDIX

Receiver code

```
#include <LiquidCrystal.h>
#include <stdio.h>
#include <string.h>
#include <Wire.h>
#include "dht.h"
LiquidCrystal lcd(6, 7, 5, 4, 3, 2);
#define dht_apin 8
dht DHT;
int co  = 9;
int vib = 10;
int buzzer = 13;
int tempc=0,humc=0;
int ADXL345 = 0x53;
float X_out, Y_out, Z_out;
int sti=0;
String inputString = "";    // a string to hold incoming data
boolean stringComplete = false; // whether the string is complete
void adxl_345_init()
{
    Wire.begin(); // Initiate the Wire library
    // Set ADXL345 in measuring mode
    Wire.beginTransmission(ADXL345); // Start communicating with the device
    Wire.write(0x2D); // Access/ talk to POWER_CTL Register - 0x2D
    // Enable measurement
    Wire.write(8); // (8dec -> 0000 1000 binary) Bit D3 High for measuring enable
    Wire.endTransmission();
    delay(10);
}
```

```

    }

    void adxl_345_read()
    {
        // === Read accelerometer data === //

        Wire.beginTransmission(ADXL345);
        Wire.write(0x32); // Start with register 0x32 (ACCEL_XOUT_H)
        Wire.endTransmission(false);

        Wire.requestFrom(ADXL345, 6, true); // Read 6 registers total, each axis value is
        stored in 2 registers

        X_out = ( Wire.read() | Wire.read() << 8); // X-axis value

        //X_out = X_out/256; //For a range of +-2g, we need to divide the raw values by 256,
        according to the datasheet

        Y_out = ( Wire.read() | Wire.read() << 8); // Y-axis value
        //Y_out = Y_out/256;

        Z_out = ( Wire.read() | Wire.read() << 8); // Z-axis value
        //Z_out = Z_out/256;

        /*
        Serial.print("Xa= ");
        Serial.print(X_out);
        Serial.print(" Ya= ");
        Serial.print(Y_out);
        Serial.print(" Za= ");
        Serial.println(Z_out);
        */
    }

    void setup()
    {
        Serial.begin(9600);
        serialEvent();
        pinMode(co, INPUT); pinMode(vib, INPUT_PULLUP);
        pinMode(buzzer, OUTPUT);
    }

```

```

digitalWrite(buzzer, HIGH);
lcd.begin(16, 2);lcd.cursor();
lcd.print("Coalmine Helmet");
delay(2000);
    adxl_345_init();
lcd.clear();
lcd.write("T:"); //2-3-4,0
lcd.setCursor(5,0);
lcd.write("H:"); //7-8-9,0
lcd.setCursor(10,0);
lcd.write("M:"); //12-13-14,0
lcd.setCursor(0,1);
lcd.write("C:"); //2,1
lcd.setCursor(8,1);
lcd.write("V:"); //10,1
serialEvent();
}
int memsv=0;
void loop()
{
    DHT.read11(dht_apin);
    tempc = DHT.temperature;
    humc = DHT.humidity;
    lcd.setCursor(2,0);convertl(tempc);
    lcd.setCursor(7,0);convertl(humc);
    adxl_345_read();
    if((X_out > -200 && X_out < 200) && (Y_out > -200 && Y_out < 200))
    {
        lcd.setCursor(12,0);

```

```

    lcd.print("Stb ");
    memsv=1;
}
else
{
    lcd.setCursor(12,0);
    lcd.print("Fall");
    memsv=0;
}
Serial.write('*');
converts(tempc);
converts(humc);
if(memsv == 1)
{
    Serial.write('s');
}
if(memsv == 0)
{
    Serial.write('f');
}
if(digitalRead(co) == LOW)
{
    lcd.setCursor(2,1);lcd.write("ON ");
    Serial.write('1');
    goto mn1;
}
if(digitalRead(co) == HIGH)
{
    lcd.setCursor(2,1);lcd.write("OFF");
    Serial.write('2');
}

```



```

    }
mn1:
if(digitalRead(vib) == LOW)
{
    lcd.setCursor(10,1);lcd.write("ON ");
    Serial.write('3');
    goto mn2;
}
if(digitalRead(vib) == HIGH)
{
    lcd.setCursor(10,1);lcd.write("OFF");
    Serial.write('4');
}
mn2: Serial.write('#');
    if(tempc > 40 || humc > 90 || memsv == 0 || digitalRead(co) == LOW ||
digitalRead(vib) == LOW)
    {
        digitalWrite(buzzer, LOW);
    }
else
    {
        digitalWrite(buzzer, HIGH);
    }
    delay(2000);
}
void serialEvent()
{
    while(Serial.available())
    {
        char inChar = (char)Serial.read();
    }
}

```

```

    }
    /*
void serialEvent()
{
    while(Serial.available())
    {
        char inChar = (char)Serial.read();
        if(inChar == '*')
        {
            sti=1;
        }
        if(sti == 1)
        {
            inputString += inChar;
        }
        if(inChar == '#')
        {
            sti=0;
            stringComplete = true;
        }
    }
}
*/

void converts(unsigned int value)
{
    unsigned int a,b,c,d,e,f,g,h;

    a=value/10000;
    b=value%10000;
    c=b/1000;
    d=b%1000;
    e=d/100;

```

```

        f=d%100;
        g=f/10;
        h=f%10;
        a=a|0x30;
        c=c|0x30;
        e=e|0x30;
        g=g|0x30;
        h=h|0x30;
        //Serial.write(a);
        //Serial.write(c);
        Serial.write(e);
        Serial.write(g);
        Serial.write(h);
    }
    void convertl(unsigned int value)
    {
        unsigned int a,b,c,d,e,f,g,h;
        a=value/10000;
        b=value%10000;
        c=b/1000;
        d=b%1000;
        e=d/100;
        f=d%100;
        g=f/10;
        h=f%10;
        a=a|0x30;
        c=c|0x30;
        e=e|0x30;
        g=g|0x30;
        h=h|0x30;
    }

```

```

        //lcd.write(a);

        //lcd.write(c);

        lcd.write(e);

        lcd.write(g);

        lcd.write(h);

    }

```

Transmitter code

```

#include <LiquidCrystal.h>

#include <stdio.h>

//LiquidCrystal lcd(6, 7, 5, 4, 3, 2); //rs,en,d4,d5,d6,d7
LiquidCrystal lcd(13, 12, 14, 27, 26, 25);

#include <WiFi.h>

#include <HTTPClient.h>

HTTPClient http;

const char *ssid = "iotserver";

const char *password = "iotserver123";

int httpResponseCode;

String servername = "http://projectsfactoryserver.in/storeddata.php?name=";

String accountname = "iot1221";

String field1 = "&s1=";

String field2 = "&s2=";

String field3 = "&s3=";

String field4 = "&s4=";

String field5 = "&s5=";

String payload="";

String co_string="";

String vib_string="";

String mems_string="";

char gchr='x';

```

```

int sti=0;

String inputString = "";    // a string to hold incoming data
boolean stringComplete = false; // whether the string is complete
int cntlmk=0;
int tempc=0,humc=0;
char memsv='x';
char coss='x';
char vibs='x';
int cntlmk1=0;
char rcv,pastnumber[11];
int buzzer = 23;

void beep()
{
    digitalWrite(buzzer, LOW);delay(2000); digitalWrite(buzzer, HIGH);delay(200);
}

void okcheck0()
{
    unsigned char rcr;
    do{
        rcr = Serial.read();
    }while(rcr != 'K');
}

String fire1_string="";
String fire2_string="";
String fire3_string="";

void iot_send()
{
    lcd.setCursor(15,0);lcd.print("U");

    http.begin(servername + accountname + field1 + String(tempc) + field2 + String(humc) +
field3 + co_string + field4 + vib_string + field5 + mems_string);

    httpStatusCode = http.GET();
}

```

```

    if(httpResponseCode>0)
    {
        payload="";
        //Serial.print("HTTP Response code: ");
        //Serial.println(httpResponseCode);
        payload = http.getString();
        //Serial.println(payload);
    }
    else
    {
        ;
        //Serial.print("Error code: ");
        //Serial.println(httpResponseCode);
    }
    delay(3000);
    lcd.setCursor(15,0);lcd.print(" ");
}

void setup()
{
    Serial.begin(9600);
    pinMode(buzzer, OUTPUT);
    digitalWrite(buzzer, HIGH);
    lcd.begin(16, 2);lcd.cursor();
    lcd.print("  welcome GSR ");
    lcd.setCursor(0,1);
    lcd.print("  Sir ");
    delay(2500);
    WiFi.begin(ssid, password);
    //Serial.println("Connecting");
    while(WiFi.status() != WL_CONNECTED)

```

```

    {
        delay(500);
        //Serial.print(".");
    }
    lcd.clear();
    lcd.write("T:"); //2-3-4,0
    lcd.setCursor(5,0);
    lcd.write("H:"); //7-8-9,0
    lcd.setCursor(10,0);
    lcd.write("M:"); //12-13-14,0
    lcd.setCursor(0,1);
    lcd.write("C:"); //2,1
    lcd.setCursor(8,1);
    lcd.write("V:"); //10,1
    // serialEvent();
}
void loop()
{
    if(stringComplete)
    {
        tempc = (((inputString[1]-48)*100) + ((inputString[2]-48)*10) + (inputString[3]-48));
        humc = (((inputString[4]-48)*100) + ((inputString[5]-48)*10) + (inputString[6]-48));
        memsv = inputString[7];
        coss = inputString[8];
        vibs = inputString[9];

        lcd.setCursor(2,0);convertl(tempc);
        lcd.setCursor(7,0);convertl(humc);
        mems_string="";
        if(memsv == 's')

```

```

    {
        lcd.setCursor(12,0);lcd.print("Stb");
        mems_string="Stable";
    }
if(memsv == 'f')
    {
        lcd.setCursor(12,0);lcd.print("Fal");
        mems_string="Fall";
    }

co_string="";
if(coss == '1')
    {
        lcd.setCursor(2,1);lcd.print("ON ");
        co_string="ON";
    }
if(coss == '2')
    {
        lcd.setCursor(2,1);lcd.print("OFF");
        co_string="OFF";
    }

vib_string="";
if(vibs == '3')
    {
        lcd.setCursor(10,1);lcd.print("ON ");
        vib_string="ON";
    }
if(vibs == '4')
    {

```



```

    lcd.setCursor(10,1);lcd.print("OFF");
    vib_string="OFF";
}
if(tempc > 40 || humc > 90 || memsv == 'f' || coss == '1' || vibs == '3')
{
    digitalWrite(buzzer, LOW);
}
else
{
    digitalWrite(buzzer, HIGH);
}
inputString = "";
stringComplete = false;
cntlmk1++;
if(cntlmk1 >= 10)
{
    cntlmk1=0;
    iot_send();
}
}
while(Serial.available() > 0)
{
    char chrt = (char)Serial.read();
    if(chrt == '*')
    {
        sti=1;
    }
    if(sti == 1)
    {
        inputString += chrt;
    }
}

```

```

        if(chrt == '#')
        {
            sti=0;
            stringComplete = true;
        }
    }

}

/*
void serialEvent()
{
    while(Serial.available())
    {
        char inChar = (char)Serial.read();
        //sti++;
        //inputString += inChar;
        if(inChar == '*')
        {
            sti=1;
            inputString += inChar;
            // stringComplete = true;
            // gchr = inputString[sti-1]
        }
        if(sti == 1)
        {
            inputString += inChar;
        }
        if(inChar == '#')
        {
            sti=0;
            stringComplete = true;
        }
    }
}

```

```

    }
}
*/
int readSerial(char result[])
{
    int i = 0;
    while (1)
    {
        while (Serial.available() > 0)
        {
            char inChar = Serial.read();
            if (inChar == '\n')
            {
                result[i] = '\0';
                Serial.flush();
                return 0;
            }
            if (inChar != '\r')
            {
                result[i] = inChar;
                i++;
            }
        }
    }
}

void gsmininit()
{
    Serial.write("AT\r\n");          okcheck0();
    Serial.write("ATE0\r\n");        okcheck0();
    Serial.write("AT+CMGF=1\r\n");    okcheck0();
}

```

```

Serial.write("AT+CNMI=1,2,0,0\r\n");    okcheck0();
Serial.write("AT+CSMP=17,167,0,0\r\n"); okcheck0();
lcd.clear();
lcd.print("SEND MSG STORE");
lcd.setCursor(0,1);
lcd.print("MOBILE NUMBER");
do{
    rcv = Serial.read();
}while(rcv != '*');
    readSerial(pastnumber);
lcd.clear();
lcd.print(pastnumber);
pastnumber[10]='\0';
    delay(4000); delay(4000);
    Serial.write("AT+CMGS=\"");
    Serial.write(pastnumber);
    Serial.write("\r\n"); delay(3000);
    Serial.write("Mobile no. registered\r\n");
    Serial.write(0x1A);    delay(4000); delay(4000);
//delay(1000);
}
void converts(unsigned int value){
    unsigned int a,b,c,d,e,f,g,h;
    a=value/10000;
    b=value%10000;
    c=b/1000;
    d=b%1000;
    e=d/100;
    f=d%100;
    g=f/10;

```

```

    h=f%10;
    a=a|0x30;
    c=c|0x30;
    e=e|0x30;
    g=g|0x30;
    h=h|0x30;
    //Serial.write(a);
    //Serial.write(c);
    Serial.write(e);
    Serial.write(g);
    Serial.write(h);
}
void convertl(unsigned int value)
{
    unsigned int a,b,c,d,e,f,g,h;
    a=value/10000;
    b=value%10000;
    c=b/1000;
    d=b%1000;
    e=d/100;
    f=d%100;
    g=f/10;
    h=f%10;
    a=a|0x30;
    c=c|0x30;
    e=e|0x30;
    g=g|0x30;
    h=h|0x30;
    //lcd.write(a);
    //lcd.write(c);

```

```
lcd.write(e);  
lcd.write(g);  
lcd.write(h);}
```